

Differentially Private Market Segmentation*

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Abstract

Access to consumer data enables firms to segment markets and charge different prices, but the use of such data raises privacy concerns. This paper studies a monopoly market in which information used for pricing must satisfy differential privacy, a restriction on the likelihood ratio of any disclosed message across consumer valuations. We characterize the attainable combinations of consumer and producer surplus. Relative to unrestricted disclosure, differential privacy can lower the maximum attainable consumer surplus by restricting information that supports lower prices that benefit some consumers, while the minimum attainable producer surplus remains equal to the uniform monopoly profit. Sufficiently strict privacy permits only uniform monopoly pricing, eliminating consumers' gains from segmentation. Protecting consumer data therefore need not improve consumer welfare when we consider only its effects on market outcomes.

Keywords: Differential privacy; market segmentation; price discrimination; information design; welfare.

JEL classification: D42, D82, D83, L51.

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1. Introduction

Firms use consumer data to decide which customers to charge more and which to offer a discount. Some consumers end up paying more, while others receive lower prices or buy a product they would otherwise have forgone. [Shiller \(2020\)](#) finds that web-browsing histories substantially increase the potential profit gains from price discrimination relative to demographics alone. In a field experiment, [Dubé and Misra \(2023\)](#) find that personalization raises expected profits by 19% relative to optimized uniform pricing. More than 60% of consumers benefit, yet aggregate consumer surplus falls. These findings suggest that the welfare consequences of consumer information depend on how it changes the prices offered to different buyers.

Privacy regulation constrains the information on which these prices can depend. The European Union's General Data Protection Regulation (GDPR) and the California Privacy Rights Act (CPRA) are prominent efforts to limit businesses' use of personal information and strengthen consumers' control over their data.¹ Restricting what a firm learns can prevent it from identifying buyers who would pay a higher price. It can also prevent the firm from identifying groups to whom a discount would be profitable. How much can consumers gain or lose from price discrimination when disclosure is restricted? What does attaining either extreme require of the firm's profit?

We study these questions in a monopoly market in which disclosure must satisfy differential privacy. We characterize the attainable pairs of consumer and producer surplus across all disclosure rules that meet a given privacy guarantee. We show that, at any given privacy level, every segmentation that maximizes consumer surplus leaves the firm earning exactly what it would earn under uniform monopoly price.

In our model, an information designer observes each consumer's valuation and commits to a rule that sends a report to the firm. The firm then chooses a price for consumers according to the observed report. We restrict what these reports can reveal through differential privacy

¹Article 5(1)(b)–(c) of [Regulation \(EU\) 2016/679 \(General Data Protection Regulation\)](#) establishes purpose limitation and data minimization. The CCPA, as amended by the CPRA, includes rights to correct inaccurate personal information and to limit the use and disclosure of sensitive personal information; see the [California Attorney General's explanation](#).

(Dwork et al., 2006). The restriction applies directly to the rule mapping an individual consumer’s valuation into a report, as in local differential privacy (Kasiviswanathan et al., 2011): the probability of any given report can differ across valuations by at most a factor e^ϵ .² A report can make a high valuation more likely without revealing it with certainty. Our analysis concerns the resulting prices and surplus.

The firm can always keep charging its uniform monopoly price after every report, so disclosure cannot reduce expected profit below the uniform monopoly benchmark. We show that every segmentation maximizing consumer surplus leaves the firm indifferent between the price charged in each segment and the uniform monopoly price. In segments receiving a discount, revenue from new buyers exactly offsets the lower payments of consumers who would have bought anyway. Privacy restricts which consumers can be grouped together, but this indifference continues to hold.

Whether disclosure can support a discount depends on the level of privacy guarantee. Under sufficiently strict privacy, the original uniform price remains uniquely optimal after every report, so disclosure changes neither prices nor welfare. Relaxing privacy allows other prices to become optimal in some segments. Prices below the uniform monopoly price benefit consumers; while prices above it induce losses. We call the smallest privacy budget at which a price becomes optimal in some segment its price-activation threshold. Gains or losses can arise at the threshold itself, since the firm may choose that price even when it ties with another. With two valuations, the possibilities depend on the original uniform price. If it is high, discounts allow low-valuation consumers to buy and reduce payments for high-valuation consumers who receive them. If it is low, everyone already buys, and disclosure can lead some consumers to face a higher price.

To characterize the minimum level of consumer surplus, we consider the demand that supports these higher prices. When the firm raises its price, it loses some consumers with low valuations and relies on those with sufficiently high valuations to keep buying. Our upper-tail revenue

²Local differential privacy has applications in consumer-data collection by Google, Apple, and Microsoft (Erlingsson, Pihur, and Korolova, 2014; Apple, 2017; Ding, Kulkarni, and Yekhanin, 2017). Differential privacy also protects the published tabulations of the 2020 U.S. Census (Abowd et al., 2022).

bound concerns demand from this latter group. It limits the revenue these consumers can support relative to the firm’s uniform monopoly profit. Under this condition, we show that some segmentation minimizes consumer surplus while leaving the firm’s profit unchanged. Consumers may pay more in some segments, but the additional payments are offset by lost revenue elsewhere. The aggregate decline in consumer surplus then equals the loss of total surplus from trades that no longer take place.

These pricing incentives also give a finite characterization of the full set of welfare outcomes. Reports that induce the same price can be pooled without changing either surplus or violating privacy, so each price needs at most one segment. With finitely many valuations, the attainable surplus pairs form a polygon. In the binary case, the set is a point below the alternative price’s activation threshold, a line segment at that threshold, and a triangle above it. Relaxing privacy restriction weakly expands the set and can permit larger gains for consumers and the firm, as well as larger losses for consumers. Stronger privacy consequently limits both the benefits and the harms that disclosure can produce.

Related literature. Our analysis builds on information design ([Kamenica and Gentzkow, 2011](#); [Bergemann and Morris, 2019](#)) and the market-segmentation framework of [Bergemann, Brooks, and Morris \(2015\)](#). They characterize the welfare triangle attainable through unrestricted segmentation of a single-product monopoly. Subsequent work extends this analysis to multiple products ([Haghpannah and Siegel, 2022](#)) and competition ([Elliott et al., 2026](#); [Rhodes and Zhou, 2024](#)). We keep the single-product monopoly setting and limit what the firm can learn about consumers through segmentation. This allows us to ask how surplus are redistributed under privacy regulation and how the firm’s pricing incentives determine their limits.

The closest paper is [Fallah et al. \(2024\)](#), who also introduce privacy into the environment of [Bergemann, Brooks, and Morris \(2015\)](#) (BBM hereafter). They model privacy guarantee using a market-level randomized response mechanism: with some probability, the mechanism outputs a market uniformly drawn from the set of all markets instead of the true market. In our paper, the designer can choose any disclosure channel for individual valuations satisfying local differential privacy, without imposing a particular form of noise. There are no market-level

distortions; only the disclosure channel is restricted. The firm can therefore secure its uniform monopoly profit by ignoring reports, allowing us to study how privacy changes attainable welfare while preserving this profit benchmark.

A related literature studies how the incentives of consumers and intermediaries determine the information that reaches firms. [Taylor \(2004\)](#) examines the sale of customer information; [Ichihashi \(2020\)](#), [Hidir and Vellodi \(2021\)](#), and [Ali, Lewis, and Vasserman \(2023\)](#) give consumers control over disclosure; and [Bonatti and Cisternas \(2020\)](#) study the scores intermediaries construct from consumer histories. We instead consider a fully committed designer to choose any disclosure satisfying an exogenous privacy constraint. Characterizing all such choices separates what the guarantee permits from the incentives that select a particular disclosure.³

Our formulation of the privacy constraint is based on [Schmutte and Yoder \(2024\)](#), who express differential privacy as a support restriction on Bayes-plausible posteriors for statistical publication. [Pan et al. \(2025\)](#) study Bayesian persuasion under several privacy notions. We use the posterior approach to characterize consumer and producer surplus jointly. The all-pairs likelihood-ratio constraint generates two-level extreme points related to the staircase mechanisms of [Kairouz, Oh, and Viswanath \(2016\)](#).⁴

Sections 2 and 3 establish the settings and the uniform monopoly price benchmarks. Section 4 illustrates the pricing and welfare effects in the binary model. Section 5 characterizes the welfare polygon, consumer-surplus extrema, and price-activation thresholds for finite valuations. Section 6 concludes. Appendix A contains the supporting results and proofs.

³Work on differentially private dynamic pricing studies the revenue cost of privacy for learning algorithms (e.g., [Chen, Miao, and Wang, 2023](#)). [Acquisti, Taylor, and Wagman \(2016\)](#) survey the broader economics of privacy.

⁴Other approaches include the social choice treatment of accuracy and privacy in [Abowd and Schmutte \(2019\)](#), the posterior-based privacy measure in [Eilat, Eliaz, and Mu \(2021\)](#), and the exact non-disclosure of designated events in [Strack and Yang \(2024\)](#).

2. Model

2.1 Environment and Market Segmentation

A monopolistic firm produces a single product at zero marginal cost. Information about consumers allows the firm to offer different prices to different groups; the privacy constraint will limit how those groups can be formed. There is a continuum of consumers with total mass one; each demands at most one unit and knows her own valuation. Valuations lie in the finite set

$$V = \{v_1, v_2, \dots, v_K\}, \quad K \geq 2,$$

where

$$0 < v_1 < v_2 < \dots < v_K < \infty.$$

A consumer who buys at price p obtains utility $v - p$; one who does not buy obtains zero.⁵

A *market* is a probability distribution x over V , with x_k the share of consumers whose valuation is v_k . The set of all markets is

$$X := \Delta(V) = \left\{ x \in \mathbb{R}_+^K : \sum_{k=1}^K x_k = 1 \right\}.$$

An exogenously given aggregate market x^* has full support, $x_k^* > 0$ for every k , so every valuation occurs before any information is disclosed. This also keeps the posterior-to-prior likelihood ratios used below well-defined.

A *segmentation* σ divides the aggregate market into finitely many submarkets. Formally, σ is a finite-support probability distribution over X , with $\sigma(x)$ the share of consumers assigned to market x , and the feasible segmentations are those that average back to the aggregate market:

$$\Sigma := \left\{ \sigma \in \Delta(X) : \sum_{x \in \text{supp}(\sigma)} \sigma(x) \cdot x = x^*, |\text{supp}(\sigma)| < \infty \right\}.$$

⁵The setting can equivalently be formulated as a single-consumer problem in which the consumer's valuation is uncertain.

2.2 Privacy-Constrained Information

An information designer observes each consumer's valuation and commits to an information structure (M, ψ) : a finite message space M , consistent with the finite-support segmentations in Σ , and a stochastic disclosure rule

$$\psi : V \rightarrow \Delta(M)$$

that maps each valuation to a distribution over messages. Conditional on valuation v_k , a message is drawn from $\psi(\cdot | v_k)$ and disclosed to the firm. Without loss of generality, every message is sent with positive probability under some valuation.⁶

The disclosure must protect the consumer's valuation. For an exogenously given privacy budget $\epsilon \geq 0$, the channel must satisfy ϵ -*differential privacy*.

Definition 1 (Differential privacy (Dwork et al., 2006; Kasiviswanathan et al., 2011)). *An information structure is ϵ -differentially private (ϵ -DP) if for every $m \in M$ and every pair $v_k, v_{k'} \in V$ with $k \neq k'$,*

$$\left| \log \frac{\psi(m | v_{k'})}{\psi(m | v_k)} \right| \leq \epsilon.$$

Definition 1 requires every message to remain possible under every valuation. A message may make a high valuation more likely, but it cannot identify a consumer as a high type with certainty. The likelihood bound is equivalent to

$$\psi(m | v_{k'}) \leq e^\epsilon \psi(m | v_k) \quad \text{for every } m, k, \text{ and } k'.$$

The factor e^ϵ limits what any report can teach the firm: whatever the message, the firm's odds between any two valuations move by at most a factor e^ϵ relative to its prior. With $\epsilon = \log 2$, for example, no message can more than double the odds that a consumer's valuation is high rather than low. The restriction is imposed on the channel itself and does not depend on the aggregate market x^* .

⁶A message that is sent with probability zero under every valuation has no effect on posterior beliefs or market outcomes and can therefore be deleted.

The constraint in Definition 1 is ϵ -local differential privacy (LDP) on a finite type space (Kasiviswanathan et al., 2011): each consumer's valuation is privatized separately before her report reaches the firm. The mechanism descends from randomized response (Warner, 1965) and is the guarantee under which large technology firms collect consumer data—Google's RAPPOR, Apple's device analytics, and Microsoft's telemetry (Erlingsson, Pihur, and Korolova, 2014; Apple, 2017; Ding, Kulkarni, and Yekhanin, 2017). The designer can be interpreted as a trusted intermediary: it observes the consumer's true valuation and release only the privatized message. LDP limits what the firm can infer from the report used to set the consumer's price before purchase.⁷ It does not protect the consumer against the intermediary itself.

Consumers who generate the same message form a segment. After observing that message, the firm's posterior distribution is the distribution of valuations within the segment. By Bayes' rule, its odds satisfy

$$\frac{x_{k'}}{x_k} = \frac{\psi(m|v_{k'})}{\psi(m|v_k)} \frac{x_{k'}^*}{x_k^*}.$$

A message that is more likely under $v_{k'}$ than under v_k raises the relative share of $v_{k'}$ consumers in its segment. Differential privacy bounds precisely this change.

Therefore, (M, ψ) is ϵ -DP if and only if every induced segment's log odds between any two valuations stay within ϵ of the aggregate market's:

$$\left| \log \frac{x_{k'}}{x_k} - \log \frac{x_{k'}^*}{x_k^*} \right| \leq \epsilon, \quad \text{for every } k \neq k'. \quad (1)$$

For the fixed aggregate market x^* , let

$$X(\epsilon) := \{x \in X : x \text{ satisfies (1)}\}$$

denote the set of private posterior markets. Every such market has full support: even a message suggesting a high willingness to pay must sometimes come from a low-valuation consumer. At zero budget, messages reveal nothing and $X(0) = \{x^*\}$. A larger budget allows larger changes in

⁷LDP does not cover the realized purchase, which can itself reveal information about the consumer's willingness to pay.

the odds, so the feasible set expands (Lemma A.1).

The connection between channels and segmentations also runs backward. Every Bayes-plausible segmentation supported on $X(\epsilon)$ can be generated by a private channel. To see the construction, use a message for each segment x and assign a type- v_k consumer to it with probability $\sigma(x)x_k/x_k^*$. The numerator is the mass of these consumers needed in that segment, and the denominator is their total mass. Bayes plausibility ensures that these assignment probabilities sum to one. Appendix A.1 verifies the privacy constraint.

Thus the designer can equivalently choose a channel or a segmentation from

$$\Sigma(\epsilon) := \{\sigma \in \Sigma : \text{supp}(\sigma) \subseteq X(\epsilon)\}.$$

We work with segmentations below because they make the firm's pricing incentives and the resulting surplus transparent. The channel construction ensures that each segmentation has an implementable disclosure rule behind it.

2.3 Pricing and Welfare

The timing is as follows. The designer commits to an ϵ -DP information structure; a message is generated from each consumer's valuation and disclosed to the firm; the firm chooses a message-contingent price; the consumer observes her price and decides whether to buy. Consumers purchase when indifferent, so a consumer with valuation v_j buys at price p if and only if $v_j \geq p$.

Although the firm may charge any nonnegative price, restricting attention to prices in V is without loss.⁸

For each v_k and market x ,

$$R_k(x) := v_k \sum_{j=k}^K x_j$$

⁸For $k = 2, \dots, K$, a price in $(v_{k-1}, v_k]$ sells to the same consumers as v_k at weakly lower revenue. A price below v_1 is dominated by v_1 , and a price above v_K sells nothing.

is the firm's revenue from charging v_k in market x . At that price, consumers with valuations at least v_k buy, and their surplus is

$$U_k(x) := \sum_{j=k}^K (v_j - v_k) x_j.$$

Raising the price increases the margin on consumers who continue buying, but excludes those whose valuations fall below it. The composition of the segment determines which price best balances these effects. Price v_k is *optimal* in x if $R_k(x) \geq R_i(x)$ for every $i = 1, \dots, K$, and

$$X_k := \{x \in X : R_k(x) \geq R_i(x), \text{ for every } i = 1, \dots, K\}$$

denotes the set of markets in which v_k is optimal. Given the aggregate market x^* , select

$$i^* \in \operatorname{argmax}_{k \in \{1, \dots, K\}} R_k(x^*),$$

write $v^* := v_{i^*}$ for the uniform monopoly price, and let $X^* = X_{i^*}$ be its optimality region. The following regularity condition keeps the aggregate market away from a price tie.

Assumption 1. $x^* \in \operatorname{ri}_X(X^*)$, the interior of X^* relative to X .

Assumption 1 says the aggregate market lies in the interior of the region in which v^* is optimal. It rules out a tie at the no-disclosure benchmark and makes v^* unique (Lemma A.2).

Under ϵ -DP, the markets in which v_k is optimal are

$$X_k(\epsilon) := X_k \cap X(\epsilon).$$

An empty $X_k(\epsilon)$ means that no admissible report can make the firm willing to charge v_k . If the intersection is nonempty, the price is available in at least one private market. Each nonempty intersection is a polytope (Lemma A.3); its boundaries reflect either a binding privacy constraint or the firm's indifference between prices.

The following example, adapted from BBM, shows how privacy reshapes the simplex of markets.

Example 1 (Three valuations with a uniform aggregate market). The consumers have three types of valuations, $V = \{1, 2, 3\}$, with equal proportion in the aggregate market, $x^* = (1/3, 1/3, 1/3)$. The budget is $\epsilon = \log 3$: no segment may more than triple the odds between any two valuations. As in BBM, the set of all markets is represented by a two-dimensional simplex. Figure 1 shows how ϵ -DP progressively reduce the simplex. The left panel bounds the relative shares of valuations 1 and 2. The middle panel bounds the shares of 2 and 3, and the right panel imposes the remaining bound between 1 and 3. Together these restrictions leave a hexagon around the aggregate market. Its distance from the simplex's edges reflects the impossibility of ruling out any valuation. Figure 2 illustrates the monopoly-price regions: a feasible market's position inside the hexagon determines which price the firm is willing to charge. This example will also illustrate the welfare consequences of privacy in Section 5.

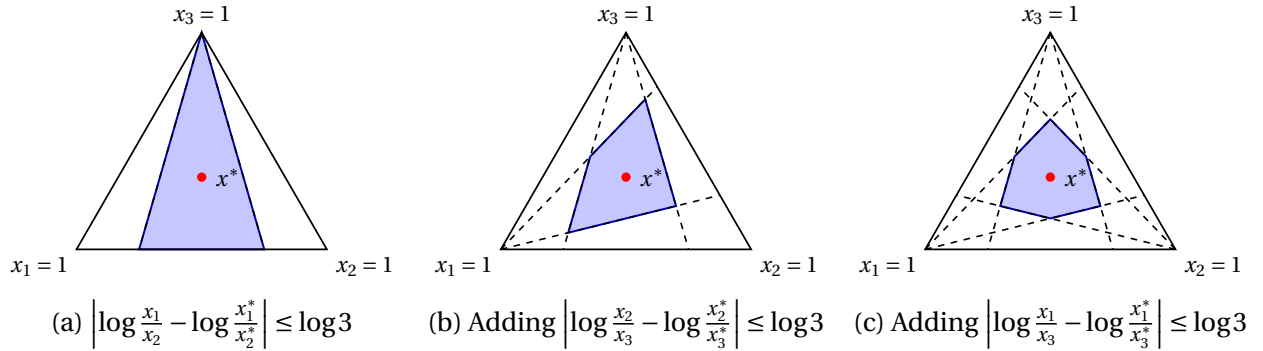


Figure 1: ϵ -Differentially Private Markets

For a segmentation σ , a *pricing rule*

$$\phi : \text{supp}(\sigma) \rightarrow \Delta(V)$$

specifies a distribution over prices in every market in the support of σ , with $\phi_k(x)$ the probability of charging v_k in market x . A pricing rule is *optimal* if

$$v_k \in \text{supp}(\phi(x)) \implies x \in X_k \quad \text{for every } x.$$

A pair (u, π) of consumer and producer surplus is a *welfare outcome*. Under the uniform monopoly

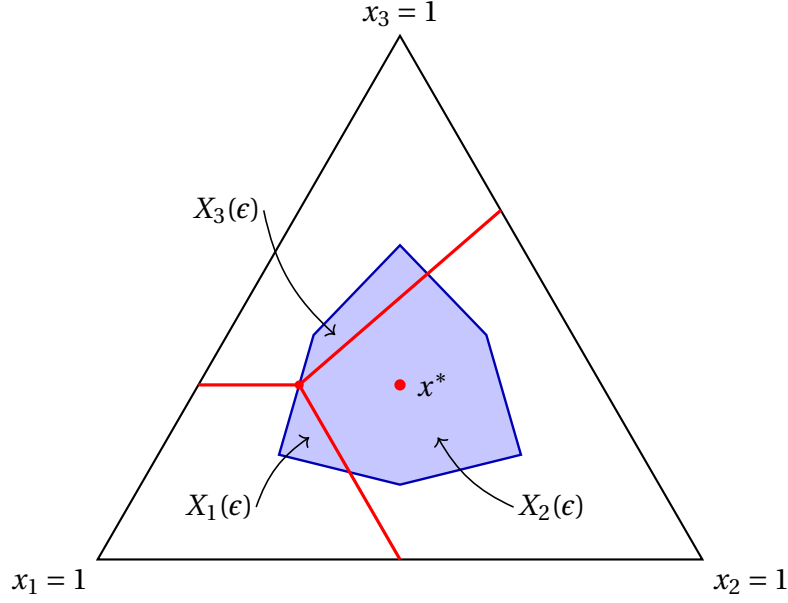


Figure 2: The Simplex of Markets

price v^* , the welfare outcome is (u^*, π^*) , where

$$u^* := \sum_{j=i^*}^K (v_j - v^*) x_j^*$$

is consumer surplus and

$$\pi^* := R_{i^*}(x^*) = v^* \sum_{j=i^*}^K x_j^*$$

is the uniform monopoly profit.

Given a segmentation σ and an associated pricing rule ϕ , consumer surplus is

$$u(\sigma, \phi) := \sum_{x \in \text{supp}(\sigma)} \sigma(x) \sum_{k=1}^K \phi_k(x) \sum_{j=k}^K (v_j - v_k) x_j,$$

producer surplus is

$$\pi(\sigma, \phi) := \sum_{x \in \text{supp}(\sigma)} \sigma(x) \sum_{k=1}^K \phi_k(x) R_k(x),$$

and total surplus is

$$w(\sigma, \phi) := \sum_{x \in \text{supp}(\sigma)} \sigma(x) \sum_{k=1}^K \phi_k(x) \sum_{j=k}^K v_j x_j.$$

In particular, $w(\sigma, \phi) = u(\sigma, \phi) + \pi(\sigma, \phi)$.

The set of attainable welfare outcomes under ϵ -DP is

$$\mathcal{W}(\epsilon) := \{(u(\sigma, \phi), \pi(\sigma, \phi)) : \sigma \in \Sigma(\epsilon), \phi \text{ is optimal}\}.$$

Without a privacy restriction, BBM show that three bounds completely describe attainable welfare. Consumers can always decline to buy, and the firm can ignore disclosure and charge v^* . Consumer surplus must therefore be nonnegative, and producer surplus cannot fall below the uniform monopoly profit. Finally, total surplus is bounded above by the value of serving every consumer.

Theorem 0 (Bergemann, Brooks, and Morris (2015)). *There exists a segmentation and associated optimal pricing rule ϕ that implement (u, π) if and only if*

$$u \geq 0, \quad \pi \geq \pi^*, \quad u + \pi \leq \sum_{j=1}^K v_j x_j^*.$$

In this paper, we do not impose a particular objective on the information designer. Our goal is the characterization of the set $\mathcal{W}(\epsilon)$, and the segmentation and pricing rules that attain its boundary. The question is how much of BBM's welfare triangle remains available once every disclosure must protect consumers' privacy.

3. Uniform-monopoly-price benchmarks

We first establish two general properties of the set of welfare outcomes under ϵ -DP. The first is inherited directly from the unrestricted BBM benchmark.

Observation 1. *For any feasible segmentation σ satisfying $\text{supp}(\sigma) \subseteq X(\epsilon)$ and any associated optimal pricing rule ϕ , the resulting producer surplus is weakly greater than π^* .*

This follows from Theorem 0, since every private segmentation is also feasible without privacy. Because $X(\epsilon)$ is a subset of the unrestricted set of markets, every segmentation satisfying ϵ -DP is also feasible in the BBM environment. Since π^* is a lower bound on producer surplus over

the larger, unrestricted set of segmentations, the same lower bound necessarily applies to the smaller set of ϵ -DP segmentations.

Although the lower bound of producer surplus is preserved under ϵ -DP, the privacy budget determines how much the firm's optimal price can vary across segments. In particular, sufficiently stringent privacy regulation makes all disclosed information irrelevant for pricing.

Proposition 1. *There exists $\underline{\epsilon} > 0$ such that, for every $\epsilon \in [0, \underline{\epsilon})$, the set of welfare outcomes is the singleton:*

$$\mathcal{W}(\epsilon) = \{(u^*, \pi^*)\}.$$

The intuition follows from the strict optimality of the uniform monopoly price. Because x^* lies in the relative interior of the set of markets in which v^* is optimal, the price v^* has a strict profit advantage over every competing price in the aggregate market. This advantage continues to hold for markets sufficiently close to x^* .

When the privacy budget is small, the ϵ -DP constraint restricts a market to be sufficiently similar to the aggregate market. Thus, all feasible markets remain within the region in which v^* is the unique optimal price. Because the firm charges v^* in every segment, price discrimination disappears. Therefore, the consumer surplus and producer surplus coincide with their no-disclosure levels, u^* and π^* .

More generally, tightening the privacy constraint reduces the set of markets, and therefore the set of attainable welfare outcomes. It limits the producer-surplus gains that may arise from segmentation, but Observation 1 shows that producer surplus cannot fall below π^* . Its effect on consumer surplus is more subtle because restricting price discrimination may affect both trade and the division of surplus. To examine these effects more closely, we first consider the case of binary valuations.

4. Binary Benchmark

With two valuations, the role of privacy regulation depends on whom the firm serves before receiving information. When $v^* = v_2$, only high-valuation consumers buy, and they keep no surplus. Information can benefit consumers by inducing the firm to offer the low price to some of them. When $v^* = v_1$, everyone already buys at the low price; information instead creates opportunities to charge some high-valuation consumers more. Privacy limits the first change but protects consumers against the second. Once privacy is relaxed enough for the firm to strictly prefer the alternative price in some segment, the set of welfare outcomes form a triangle. Its three vertices correspond to maximum producer surplus, maximum consumer surplus, and minimum consumer surplus.

Let $V = \{v_1, v_2\}$ and write $\alpha = x_1^*/(1 - x_1^*)$, so $x_1^* = \alpha/(1 + \alpha)$. A market is described by its share x_1 of low-valuation consumers. Condition (1) implies

$$X(\epsilon) = \left\{ (x_1, 1 - x_1) : \frac{\alpha}{\alpha + e^\epsilon} \leq x_1 \leq \frac{\alpha e^\epsilon}{1 + \alpha e^\epsilon} \right\}.$$

The lower endpoint is the market most concentrated on high valuations that privacy permits; the upper endpoint is the market most concentrated on low valuations. A larger budget moves the endpoints farther apart, but both types remain present at every finite budget.

The firm faces a trade-off between the revenue from serving more consumers and charging higher price to those willing to pay more. It is indifferent between charging v_1 to everyone and charging v_2 to high-valuation consumers at $x_1 = (v_2 - v_1)/v_2$. Each price is uniquely optimal in some feasible market exactly when this indifference market lies strictly inside the privacy interval:

$$\frac{\alpha}{\alpha + e^\epsilon} < \frac{v_2 - v_1}{v_2} < \frac{\alpha e^\epsilon}{1 + \alpha e^\epsilon} \iff \epsilon > \epsilon_B := \left\lceil \log \frac{v_2 - v_1}{\alpha v_1} \right\rceil. \quad (2)$$

Below ϵ_B , every feasible segment has the same optimal price. At ϵ_B , the alternative price becomes optimal at one endpoint of the interval. The firm is then indifferent, but consumers are not: at the indifference market, they receive zero if the firm charges v_2 and $v_1(v_2 - v_1)/v_2$ if it charges v_1 . The low price gives high-valuation consumers a discount and brings low-valuation

consumers into the market. The additional sales exactly compensate the firm for the lower markup. Thus welfare can change as soon as the second price becomes optimal, even though the firm need not gain any profit.

Figure 3 depicts producer surplus $\pi(x_1) = \max\{v_1, v_2(1 - x_1)\}$. The blue portion is its restriction to $X(\epsilon)$; the red chord is its concave closure. At the aggregate share of low valuations, the height of this chord gives the largest average profit that a feasible segmentation can deliver.

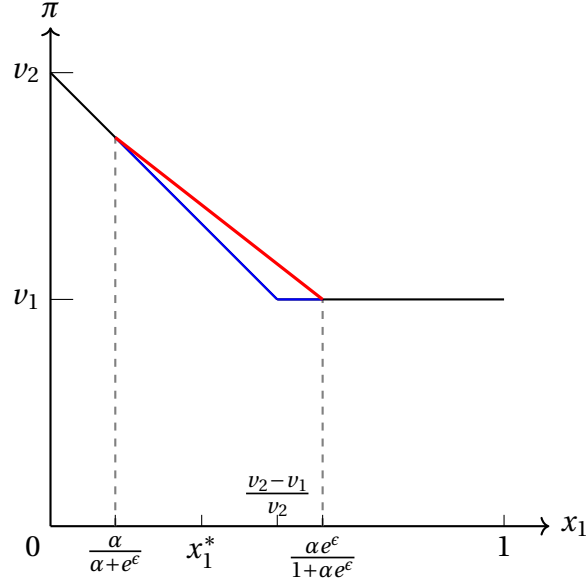


Figure 3: Producer surplus under ϵ -DP

The three boundary outcomes below are defined for $\epsilon \geq \epsilon_B$. Above the threshold, the firm strictly prefers the high price at the lower endpoint of the privacy interval and the low price at the upper endpoint. Choosing the two segments and their weights then determines how many consumers receive each price.

Maximum producer surplus. The firm gains most from separating the two types as much as privacy permits. The segment concentrated on high valuations receives the high price, while the other segment receives the low price. Notice that producer surplus is convex in x_1 , so splitting the aggregate market between the endpoints of the privacy interval reaches the red chord in Figure 3. The segment weights preserve the aggregate share of each valuation. This segmen-

tation yields

$$A_\epsilon = (u_A^\epsilon, \pi_A^\epsilon) = \left(\frac{v_2 - v_1}{(1 + \alpha)(1 + e^\epsilon)}, \frac{(1 + \alpha e^\epsilon)v_1 + e^\epsilon v_2}{(1 + \alpha)(1 + e^\epsilon)} \right). \quad (3)$$

Privacy prevents this segmentation from perfectly separating the two types. High-valuation consumers who remain in the low-price segment keep a positive surplus. Low-valuation consumers who remain in the segment priced at v_2 do not buy. The first effect limits how much surplus the firm captures; the second destroys gains from trade. This exclusion creates a total-surplus loss of

$$\bar{w} - w(A_\epsilon) = \frac{\alpha v_1}{(1 + \alpha)(1 + e^\epsilon)} > 0, \quad (4)$$

where $\bar{w} = (\alpha v_1 + v_2)/(1 + \alpha)$ is first-best surplus.

Consumer-surplus bounds. Minimum and maximum consumer surplus are attained on the profit floor $\pi = \pi^*$. The relevant segmentations keep the original uniform price optimal in both segments. One segment is an indifference market, so changing its price alters consumer surplus without changing the firm's revenue.

When $v^* = v_2$, low-valuation consumers are too scarce in the aggregate market to make the low price attractive. To induce a discount, the designer groups enough of them into an indifference market and assigns the remaining consumers to a segment concentrated on high valuations. Specifically, split the aggregate market between the lower endpoint $\alpha/(\alpha + e^\epsilon)$ and the indifference market $(v_2 - v_1)/v_2$. Charging the high or low price at the latter gives, respectively,

$$B_\epsilon = \left(0, \frac{v_2}{1 + \alpha} \right) \quad \text{and} \quad C_\epsilon = \left(\frac{\alpha v_1 (v_2 - v_1) (e^\epsilon - 1)}{(1 + \alpha)((v_2 - v_1)e^\epsilon - \alpha v_1)}, \frac{v_2}{1 + \alpha} \right). \quad (5)$$

No disclosure already gives consumers zero surplus, so privacy cannot raise the minimum. To attain the maximum, the designer wants as many consumers as possible to receive the low price at the indifference market. But the remaining segment must also contain some low-valuation consumers to satisfy privacy. Those consumers cannot be reassigned to support a discount elsewhere. This restriction reduces the maximum below its unrestricted value $\alpha v_1/(1 + \alpha)$.

When $v^* = v_1$, the aggregate market contains enough low-valuation consumers for the firm to

serve everyone. To reduce consumer surplus, the designer must now create a segment in which the high price becomes attractive. Combine the indifference market with the upper endpoint $\alpha e^\epsilon / (1 + \alpha e^\epsilon)$ instead. The two pricing choices give

$$B_\epsilon = \left(\frac{(v_2 - v_1)(\alpha v_1 - (v_2 - v_1))}{(1 + \alpha)(\alpha v_1 e^\epsilon - (v_2 - v_1))}, v_1 \right) \quad \text{and} \quad C_\epsilon = \left(\frac{v_2 - v_1}{1 + \alpha}, v_1 \right). \quad (6)$$

Privacy prevents complete separation of the two types: some high-valuation consumers must remain in the segment charging v_1 . These consumers retain surplus $v_2 - v_1$, so minimum consumer surplus remains strictly positive. Maximum consumer surplus is unaffected. Charging v_1 to everyone gives C_ϵ , where high-valuation consumers receive the largest possible surplus and everyone buys. This achieves the same consumer maximum as without privacy, along with first-best total surplus. When identifying these endpoints, we allow the firm to choose either price whenever it is indifferent.

Figures 4 and 5 plot the upper and lower envelopes of consumer surplus in the two cases.

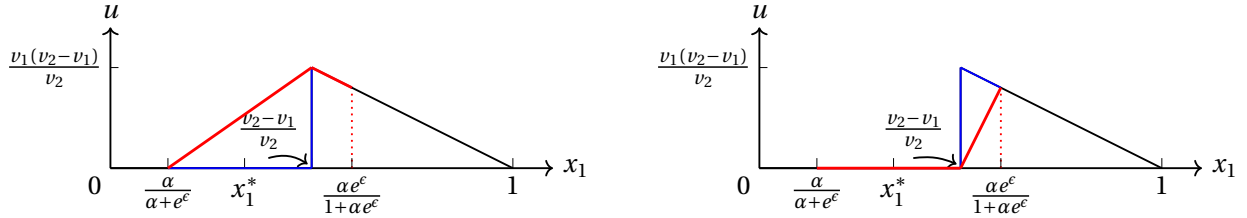


Figure 4: Maximum and minimum consumer surplus when $x_1^* < \frac{v_2 - v_1}{v_2}$

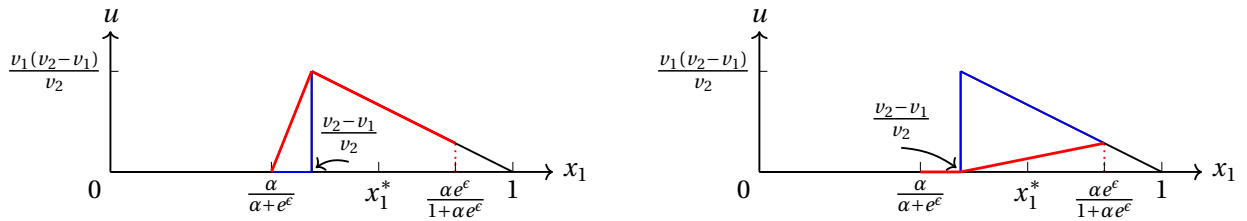


Figure 5: Maximum and minimum consumer surplus when $x_1^* > \frac{v_2 - v_1}{v_2}$

The three outcomes also describe every trade-off between consumer and producer surplus. Combining feasible segmentations fills their convex hull; the linear surplus formulas within each price region imply that no feasible outcome lies outside it. At the activation threshold,

the firm can vary consumer surplus by choosing between tied prices, but cannot yet earn more than the uniform monopoly profit.

Proposition 2 (Binary set of welfare outcomes). *If $K = 2$, then*

$$\mathcal{W}(\epsilon) = \begin{cases} \{(u^*, \pi^*)\}, & \epsilon < \epsilon_B, \\ \text{co}\{B_\epsilon, C_\epsilon\}, & \epsilon = \epsilon_B, \\ \text{co}\{A_\epsilon, B_\epsilon, C_\epsilon\}, & \epsilon > \epsilon_B, \end{cases}$$

where the vertices are given in equations (3), (5), and (6). At ϵ_B , $A_\epsilon = C_\epsilon$ if $v^* = v_2$, and $A_\epsilon = B_\epsilon$ if $v^* = v_1$.

Above the threshold, A_ϵ maximizes producer surplus, B_ϵ minimizes consumer surplus, and C_ϵ maximizes both consumer and total surplus. The last coincidence has a direct economic explanation. High-valuation consumers buy at either price, so gains in total surplus come entirely from allowing more low-valuation consumers to buy. The segmentation attaining the consumer maximum also serves the largest feasible mass of low-valuation consumers. It excludes fewer of them than the producer-optimal segmentation, and excludes none when $v^* = v_1$. Thus C_ϵ is the unique total-surplus-maximizing welfare outcome, with strictly less total-surplus loss than A_ϵ .

Figure 6 places the two private welfare triangles inside the unrestricted BBM triangle. When the uniform price is high, privacy cuts off part of the consumer gains available under unrestricted disclosure. When the uniform price is low, it prevents the most severe consumer losses. In both cases, the firm cannot extract all gains from trade: full support rules out the unrestricted producer-optimal outcome $(0, \overline{w})$. The private set of welfare outcomes is therefore a strict subset at every finite budget.

Relaxing privacy. A larger budget permits sharper separation of valuations. The firm can exploit this separation to increase profit, but its implications for consumers depend on which price prevails without disclosure.

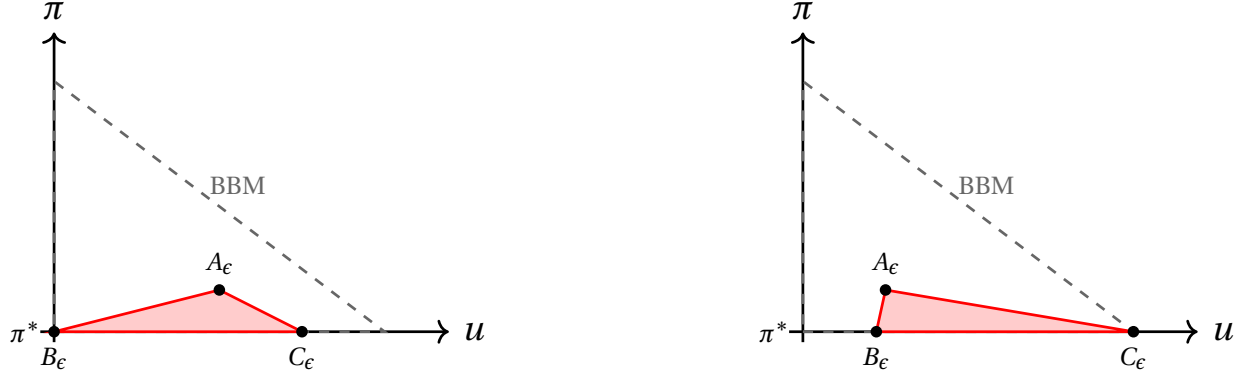


Figure 6: The binary ϵ -DP set of welfare outcomes when the uniform monopoly price is v_2 (left) and v_1 (right)

Corollary 1. *For $\epsilon > \epsilon_B$, maximum producer surplus strictly increases with ϵ . If $v^* = v_2$, maximum consumer surplus strictly increases while the minimum remains zero. If $v^* = v_1$, minimum consumer surplus strictly decreases while the maximum remains u^* .*

At the producer-optimal outcome, sharper separation sends more high-valuation consumers to the high-price segment and more low-valuation consumers to the low-price segment. The firm earns more, consumers keep less surplus, and fewer low-valuation consumers are excluded. Accordingly, A_ϵ moves up and to the left in the figures even as total surplus rises.

Along the profit floor, a larger budget permits more consumers to be assigned to the indifference market. If $v^* = v_2$, more consumers can receive the low price, so C_ϵ moves right. If $v^* = v_1$, more high-valuation consumers can be charged the high price, so B_ϵ moves left. The other end-point stays fixed because no disclosure continues to attain it. When $v^* = v_1$, maximum total surplus also remains at its first-best level. As $\epsilon \rightarrow \infty$, all three vertices converge to their unrestricted counterparts.

5. General Finite Valuations

With more than two valuations, disclosure can induce several discounts and markups relative to the uniform price. The set of welfare outcomes need not remain a triangle, but the main result for consumers survives: every segmentation that maximizes their surplus leaves the firm with

exactly π^* . Giving the firm additional profit cannot help consumers attain a higher surplus. At the other extreme, a profit-preserving segmentation attains minimum consumer surplus under a bound on the revenue that higher prices could generate.

The analysis separates the prices that privacy makes available from the welfare outcomes that the designer can achieve with them. This distinction matters because a newly available price may be used in only a small part of the market. Throughout this section, the privacy budget is finite.

5.1 A finite description of welfare outcomes

The designer may use many messages, but welfare depends on the prices those messages induce. Segments that receive the same price can be pooled: the price remains optimal, privacy is preserved, and neither surplus changes. It therefore suffices to record how many consumers of each valuation face each price.

Let q_{kj} denote the mass of valuation- v_j consumers assigned to price v_k . A feasible assignment $q = (q_{kj})$ has nonnegative entries and satisfies the following three conditions:

$$\sum_{k=1}^K q_{kj} = x_j^* \quad \text{for every } j, \quad (7)$$

$$q_{kj} x_\ell^* \leq e^\epsilon q_{k\ell} x_j^* \quad \text{for every } k, j, \ell, \quad (8)$$

$$v_k \sum_{j=k}^K q_{kj} \geq v_\ell \sum_{j=\ell}^K q_{kj} \quad \text{for every } k, \ell. \quad (9)$$

Condition (7) is Bayes plausibility, condition (8) imposes ϵ -DP within every price-indexed segment, and condition (9) makes the price indexing that segment optimal.

Write $Q(\epsilon)$ for the polytope defined by these conditions. Once consumers are assigned to prices, welfare is linear in their masses. Buyers retain the difference between valuation and price, and

the firm receives the price from everyone who buys:

$$u(q) = \sum_{k=1}^K \sum_{j=k}^K (v_j - v_k) q_{kj}, \quad (10)$$

$$\pi(q) = \sum_{k=1}^K v_k \sum_{j=k}^K q_{kj}. \quad (11)$$

Proposition 3 (The welfare polygon). *The set of attainable welfare outcomes is*

$$\mathcal{W}(\epsilon) = \{(u(q), \pi(q)) : q \in Q(\epsilon)\}. \quad (12)$$

It is a compact convex polygon, possibly a point or a line segment. Every outcome can be implemented using at most K market–price pairs, at most one per price.

The array gives both a welfare outcome and the disclosure that implements it.⁹ Convexity has a direct interpretation: randomizing between two feasible disclosures gives the corresponding average of their welfare outcomes. To find a boundary point, solve

$$\max_{q \in Q(\epsilon)} \{au(q) + b\pi(q)\}. \quad (13)$$

The weights (a, b) describe how the designer values the two surpluses; varying them traces the polygon. With several valuations, a change in these weights can alter which groups receive discounts, which face markups, and which privacy constraints bind. These additional choices explain why the boundary can have more corners than in the binary case.

Relaxing privacy enlarges $Q(\epsilon)$ and therefore $\mathcal{W}(\epsilon)$. Maximum profit, maximum consumer surplus, and maximum total surplus weakly rise, while minimum consumer surplus weakly falls. These are changes in what disclosure can achieve; they do not imply that every consumer benefits from weaker privacy. For every finite budget, the welfare polygon is a strict subset of the BBM triangle. It approaches that triangle as the privacy restriction vanishes. Appendix A.3 gives the limit argument and the geometric representation used below.

⁹A nonzero row q_k describes a segment of weight $s_k = \sum_j q_{kj}$, posterior q_k/s_k , and price v_k .

5.2 Consumer surplus and the profit floor

The profit floor provides a useful starting point for understanding how disclosure helps or hurts consumers. If the firm remains willing to charge v^* in every segment, its expected profit stays at the uniform-monopoly-price benchmark. It can still choose a different price when that price ties v^* , changing both who buys and what buyers retain.

Let $u_{\min}(\epsilon)$ and $u_{\max}(\epsilon)$ be the smallest and largest consumer surplus in $\mathcal{W}(\epsilon)$, and let $\bar{w} = \sum_j v_j x_j^*$ be first-best total surplus. Call a price *active* if it is optimal in some private market, and write

$$\mathcal{A}(\epsilon) = \{k : X_k(\epsilon) \neq \emptyset\}.$$

An active price need not be uniquely optimal; a tie is enough for the firm to charge it.

A segmentation is *profit-preserving* if v^* remains optimal in every segment:

$$\Sigma^*(\epsilon) = \{\sigma \in \Sigma(\epsilon) : \text{supp}(\sigma) \subseteq X^* \cap X(\epsilon)\}.$$

Let $u^-(\epsilon)$ and $u^+(\epsilon)$ be the minimum and maximum consumer surplus attainable by these segmentations, allowing any optimal pricing rule.

Theorem 1 (The profit floor). *A feasible segmentation yields profit π^* if and only if it belongs to $\Sigma^*(\epsilon)$. The consumer surpluses attainable on this profit floor form the interval*

$$\{u : (u, \pi^*) \in \mathcal{W}(\epsilon)\} = [u^-(\epsilon), u^+(\epsilon)].$$

At the two endpoints, the firm charges, respectively, the highest and lowest optimal price in each segment.

The firm's option to keep charging v^* explains the equivalence. Any strict improvement over this fallback in one segment raises expected profit, because no other segment can offset it with a loss. Profit therefore equals π^* precisely when the firm is indifferent between its chosen price and v^* in every segment.

This is the same indifference that supports the bottom edge of the BBM triangle. A higher tied price reduces consumer surplus through higher payments and exclusion; a lower tied price increases it through discounts and additional purchases. Privacy limits the compositions and weights of the segments that sustain these ties. The uninformative disclosure preserves profit, so

$$u_{\min}(\epsilon) \leq u^-(\epsilon) \leq u^* \leq u^+(\epsilon) \leq u_{\max}(\epsilon).$$

The interval on the profit floor could, in principle, cover only part of the attainable consumer-surplus range. At its upper endpoint, the next theorem rules out that possibility.

Theorem 2 (Consumer-optimal disclosure). *Every consumer-surplus-maximizing segmentation preserves uniform monopoly profit. Hence*

$$u_{\max}(\epsilon) = u^+(\epsilon).$$

Proof sketch. By Lemma A.7, every maximizer uses only prices at most v^* . Consumers valued at least v^* therefore buy in every segment. Collapsing them into a single type valued at v^* preserves revenues at these prices and subtracts only the constant u^* from aggregate consumer surplus. Restoring the tail in its aggregate proportions preserves privacy and keeps higher prices strictly less profitable. Lemma A.9 thus reduces the problem to a market with $v^* = v_K$, without changing producer surplus.

In this reduced market, write $p = x^*$ and $a_{rj} = q_{rj}/p_j$. Fix an arbitrary maximizer $\bar{q}$. Among maximizers with producer surplus $\pi(\bar{q})$, choose q to minimize the auxiliary objective $\mathcal{P}(q) = \sum_{r,j} p_j a_{rj}^2$. If a row earns more at its assigned price than at v_K , some lower valuation must have a larger likelihood multiplier than the highest type; otherwise, the aggregate market's strict revenue ranking would make v_K more profitable. Bayes plausibility supplies another row with the opposite multiplier inequality.

Working down the valuation tail, the proof exchanges small masses of these two types across the two rows. The changed multipliers move toward one another within each row. All-pairs privacy bounds the largest multiplier by e^ϵ times the smallest, so keeping every multiplier within its

previous range preserves every privacy inequality. Pricing incentives are checked separately: comparisons that can bind remain unchanged or improve, while adverse changes affect only comparisons with strict slack. A sufficiently small exchange therefore preserves both privacy and optimal pricing.

The exchange either raises consumer surplus or preserves both surpluses while strictly reducing $\mathcal{P}$. The first possibility contradicts consumer optimality; the second contradicts the auxiliary minimization. These contradictions rule out every positive row profit gap, so $\pi(q) = \pi^*$. Because $\pi(q) = \pi(\bar{q})$ and $\bar{q}$ was arbitrary, every maximizer preserves profit. Appendix A.4 gives the full argument. $\square$

When consumer-optimal disclosure induces a discount, revenue from additional buyers exactly offsets the lower payments of consumers who would have bought at v^* . Consumers benefit from lower prices and expanded trade while the firm retains its uniform monopoly profit.

At the opposite extreme, higher prices reduce consumer surplus through larger payments and exclusion. A minimizing disclosure never needs a price below v^* (Lemma A.7). The following condition ensures that a minimizer also preserves uniform monopoly profit.

A sufficient condition depends on the consumers who can still buy after a price increase. If no higher price is active, the uninformative disclosure attains the minimum, so $u_{\min}(\epsilon) = u^-(\epsilon) = u^*$. Otherwise, let v_a be the lowest active price above v^* and v_b the second-highest such price, setting $v_b = v_a$ when only one is active. Only consumers valued at least v_a can buy at any higher active price. Valuing this entire group at v_b gives an upper bound on what all but the highest of these prices could earn.

Proposition 4 (Minimum consumer surplus under an upper-tail bound). *Let v_a and v_b be the lowest and second-highest active prices above v^* , with $v_b = v_a$ if only one is active. If*

$$v_b \sum_{j=a}^K x_j^* \leq \pi^*,$$

then a profit-preserving segmentation attains minimum consumer surplus:

$$u_{\min}(\epsilon) = u^-(\epsilon).$$

The condition limits the ability of upper-tail demand to support higher prices relative to the profit already available under uniform pricing. It allows any number of higher active prices. The revenue bound is deliberately generous: it counts every consumer valued at least v_a as paying v_b , even if their valuation falls below that price. The highest active price is handled separately by the firm's incentives at a consumer-surplus minimum.

Consumers with valuations at least v^* and below v_a explain the first step. They buy at the benchmark price but are excluded at every higher active price. Assigning more of them to a higher-price segment makes v^* more attractive without increasing revenue at the assigned price. The proof uses small feasible reallocations of these consumers, together with the upper-tail bound, to show that any remaining profit gain would require a segment in which the highest active price is uniquely optimal. But then a small share of a lower-priced segment could be pooled into it without changing the firm's preferred price. Those consumers would retain less surplus, contradicting minimality. Thus a minimizer can be found in which every charged price ties v^* . Appendix A.4 gives the argument.

The bound holds automatically when one or two higher prices are active, since then $v_b = v_a$ and

$$v_b \sum_{j=a}^K x_j^* = R_a(x^*) < \pi^*.$$

Together with the case of no higher active price, this implies $u_{\min}(\epsilon) = u^-(\epsilon)$ whenever at most two higher prices are active. In particular, with at most three valuations, the equality holds at every finite privacy budget.¹⁰

Under the upper-tail bound, the profit floor therefore spans the entire attainable consumer-surplus range, $[u_{\min}(\epsilon), u_{\max}(\epsilon)]$. The minimum result guarantees the existence of a profit-

¹⁰For $K \geq 2$, a sufficient condition that is independent of the privacy budget is $v_{K-1} \sum_{j>i^*} x_j^* \leq \pi^*$. When at least two higher prices are active, $v_b \leq v_{K-1}$ and $a > i^*$; with fewer, the equality already holds. Conditional on buying at the uniform price, this condition requires the share of consumers valued exactly at v^* to be at least $1 - v^*/v_{K-1}$.

preserving minimizer; the maximum result applies to every maximizer. Since the former uses no price below v^* , its consumer-surplus loss relative to uniform pricing equals the loss of total surplus from forgone trades. Consumers can therefore be harmed even when the firm earns no additional profit.

Privacy also limits how far these extrema can reach. Every disclosed segment must retain every valuation, so consumers receiving different prices cannot be completely separated by valuation.

Proposition 5 (Extreme uniform prices). *For every finite privacy budget,*

$$u_{\min}(\epsilon) = 0 \iff v^* = v_K,$$

$$u_{\max}(\epsilon) = \bar{w} - \pi^* \iff v^* = v_1.$$

At any price below v_K , the highest-valuation consumers retain some surplus, and privacy places some of them in every segment. Zero consumer surplus therefore requires charging v_K everywhere. At the other extreme, any price above v_1 excludes some buyers, so first-best total surplus requires charging v_1 everywhere. Bayes plausibility permits either outcome only when the same price is optimal in the aggregate market.

It follows that maximum total surplus equals $\bar{w}$ exactly when $v^* = v_1$, and maximum profit is always strictly below $\bar{w}$. For an interior monopoly price, privacy prevents both complete surplus extraction and an efficient allocation that gives consumers all surplus above π^* .

5.3 When privacy changes consumer surplus

When privacy is sufficiently strict, every report leaves the firm charging v^* and disclosure has no welfare effect. Consumer surplus can change only once some report makes another price optimal. To locate that point, the price-activation threshold records the smallest privacy budget that makes a price available.

For a full-support market x , define

$$\rho(x) = \max_j \log \frac{x_j}{x_j^*} - \min_j \log \frac{x_j}{x_j^*}, \quad (14)$$

and set $\rho(x) = +\infty$ on the boundary of the simplex. The quantity $\rho(x)$ measures the largest difference between the log likelihoods with which two valuations generate this posterior. It is the privacy budget needed to admit that market. The privacy constraint is exactly

$$X(\epsilon) = \{x : \rho(x) \leq \epsilon\}. \quad (15)$$

For price v_k , let

$$\epsilon_k^{\text{act}} = \min_{x \in X_k} \rho(x). \quad (16)$$

These minima are attained. The uniform price has threshold zero and every other price has a positive, finite threshold, with

$$k \in \mathcal{A}(\epsilon) \iff \epsilon \geq \epsilon_k^{\text{act}}. \quad (17)$$

Appendix A.5 computes each threshold with a linear program.

The direction of the price change determines whether disclosure can benefit or harm consumers. A lower active price makes a discount possible; a higher active price makes a markup or exclusion possible. Define the first lower- and higher-price thresholds by

$$\epsilon^- = \min_{k < i^*} \epsilon_k^{\text{act}}, \quad \epsilon^+ = \min_{k > i^*} \epsilon_k^{\text{act}}, \quad (18)$$

taking the minimum of an empty set to be $+\infty$.

Proposition 6 (Activation and consumer surplus). *The consumer-surplus range first expands at the directional activation thresholds:*

$$\begin{aligned} u_{\min}(\epsilon) < u^* &\iff u^-(\epsilon) < u^* \iff \epsilon \geq \epsilon^+, \\ u_{\max}(\epsilon) > u^* &\iff \epsilon \geq \epsilon^-. \end{aligned}$$

Below the corresponding threshold, the endpoint equals u^ .*

An active price can be used in a small segment while leaving the rest of the market close enough to x^* that the firm still charges v^* . A lower price then raises the surplus of some buyers without changing the price faced by everyone else. A higher price reduces surplus in that segment through higher payments or exclusion. A tie suffices for either construction, which is why the inequalities hold at the thresholds themselves. The first gains and losses can also be achieved while preserving uniform profit.

The proposition describes the onset and direction of welfare changes. The size of those changes depends on how many consumers can be assigned to the new prices while respecting privacy and preserving the aggregate market. A larger budget can permit greater gains or losses even when the set of active prices is unchanged. Once both thresholds are reached, $u_{\min}(\epsilon) < u^* < u_{\max}(\epsilon)$: disclosure can then help consumers or hurt them, depending on its design.

5.4 Producer surplus and the rest of the polygon

To characterize maximum producer surplus, we begin with the firm's response to more detailed information. Splitting a segment into smaller segments that satisfy privacy cannot reduce expected profit: the firm can continue charging its original price in each segment or choose prices better suited to their demand. This allows us to construct a producer-optimal disclosure from the vertices of the set of private markets $X(\epsilon)$.

These vertices have a simple economic interpretation. Each places as much weight as privacy permits on a particular group of valuation types. Let S be a nonempty proper subset of $\{1, \dots, K\}$, and write $x^*(S) = \sum_{j \in S} x_j^*$ for this group's share in the aggregate market. We denote the corresponding market by $z^S(\epsilon)$. Its share of valuation- v_j consumers is

$$z_j^S(\epsilon) = \begin{cases} \frac{e^\epsilon x_j^*}{1 + (e^\epsilon - 1)x^*(S)}, & j \in S, \\ \frac{x_j^*}{1 + (e^\epsilon - 1)x^*(S)}, & j \notin S. \end{cases} \quad (19)$$

Starting from the aggregate market, this construction multiplies the shares of types in S by e^ϵ and then normalizes all shares to sum to one. It preserves relative shares within each group while increasing the odds of membership in S by the largest factor privacy allows.

For $\epsilon > 0$, these markets are exactly the vertices of $X(\epsilon)$, as established in Lemma A.5. They generalize the two endpoints of the privacy interval in the binary model. Every private market is a mixture of these vertices, so refining segments into them weakly raises profit. Maximizing producer surplus therefore reduces to choosing their weights.

Proposition 7 (Producer-optimal disclosure). *Maximum producer surplus is attained on at most K markets $z^S(\epsilon)$ and equals*

$$\max_{\lambda \geq 0} \left\{ \sum_S \lambda_S \max_k R_k(z^S(\epsilon)) : \sum_S \lambda_S z^S(\epsilon) = x^* \right\}.$$

The firm benefits weakly from replacing a pooled market with more detailed reports, because it can always ignore the distinction and keep its old price. Formally, profit is convex in the posterior: the firm chooses the best of several linear revenue functions. Splitting a segment into the privacy vertices that average to it therefore weakly raises profit. The designer can repeat this refinement until only privacy vertices remain, then choose their weights to match the aggregate market. These weights are the remaining choice in the proposition's linear program.

Let

$$\bar{\pi}_\epsilon(u) = \max \{ \pi : (u, \pi) \in \mathcal{W}(\epsilon) \}$$

be the largest profit compatible with consumer surplus u . This upper boundary records the trade-offs faced by a designer who wants to raise profit while delivering a given surplus to consumers. It is concave and piecewise linear, since $\mathcal{W}(\epsilon)$ is a polygon. Each corner marks a change in the disclosure that best balances the two objectives.

If segmentation can raise profit, the upper boundary lies strictly above π^* in the interior of the consumer-surplus range. At $u_{\max}(\epsilon)$ it returns to π^* by Theorem 2: reaching the greatest consumer surplus leaves no room for an additional profit gain. Away from the producer optimum,

boundary outcomes can require price-indifference markets as well as privacy vertices.¹¹

When the upper-tail bound in Proposition 4 holds, both consumer-surplus endpoints are attainable at π^* . Convexity then gives

$$\mathcal{W}(\epsilon) = \{(u, \pi) : u_{\min}(\epsilon) \leq u \leq u_{\max}(\epsilon), \pi^* \leq \pi \leq \bar{\pi}_\epsilon(u)\}. \quad (20)$$

In particular, this description applies whenever at most two prices above v^* are active, including every market with at most three valuations.

Example 1 continued. For $V = \{1, 2, 3\}$ and $x^* = (1/3, 1/3, 1/3)$, the uniform price is 2, with $\pi^* = 4/3$ and $u^* = 1/3$. Both other prices activate at $\epsilon = \log 2$. Below this threshold, every consumer faces price 2, so the set of welfare outcomes is a point. At the threshold, the firm can offer a discount to 1 or a markup to 3 through price ties, and the set becomes a line segment on the profit floor. Above it, disclosure can also increase profit, and the set is a pentagon.

Only one price above v^* can be active, so Proposition 4 requires no additional restriction here. Whenever price 3 is active, $v_a = v_b = 3$ and the bound holds automatically: $v_b x_3^* = 1 < 4/3 = \pi^*$. Thus both consumer-surplus endpoints lie on the profit floor, as in equation (20). At $\epsilon = \log 3$, the vertices are

$$B = \left(\frac{1}{15}, \frac{4}{3}\right), \quad C = \left(\frac{3}{5}, \frac{4}{3}\right), \quad D = \left(\frac{1}{12}, \frac{17}{12}\right), \quad D' = \left(\frac{1}{2}, \frac{17}{12}\right), \quad A = \left(\frac{4}{15}, \frac{22}{15}\right).$$

Figure 7 displays the resulting trade-offs. The endpoints B and C lie on the profit floor. At B , a share $8/15$ of consumers faces price 3, so only the highest valuation buys in that segment, retaining no surplus. At C , the same share faces price 1, so every valuation buys and buyers above 1 receive a discount. The remaining consumers face the uniform price 2. Both nonuniform prices tie 2 in their respective segments, keeping expected profit at $4/3$.

The producer-optimal point A uses privacy vertices throughout. The intermediate vertices D

¹¹Appendix A.3 describes these relative extreme points and shows that at most K market-price pairs generate any welfare vertex.

and D' show how the additional pricing option changes the boundary: D is implemented with prices 2 and 3, whereas D' uses prices 1 and 2. They give the firm the same profit but leave consumers with different surplus. The pentagon thus reflects the possibility of designing disclosure around either markups or discounts, as well as using both. Appendix A.7 gives the vertex formulas and implementing segmentations, and proves completeness through a facet description.

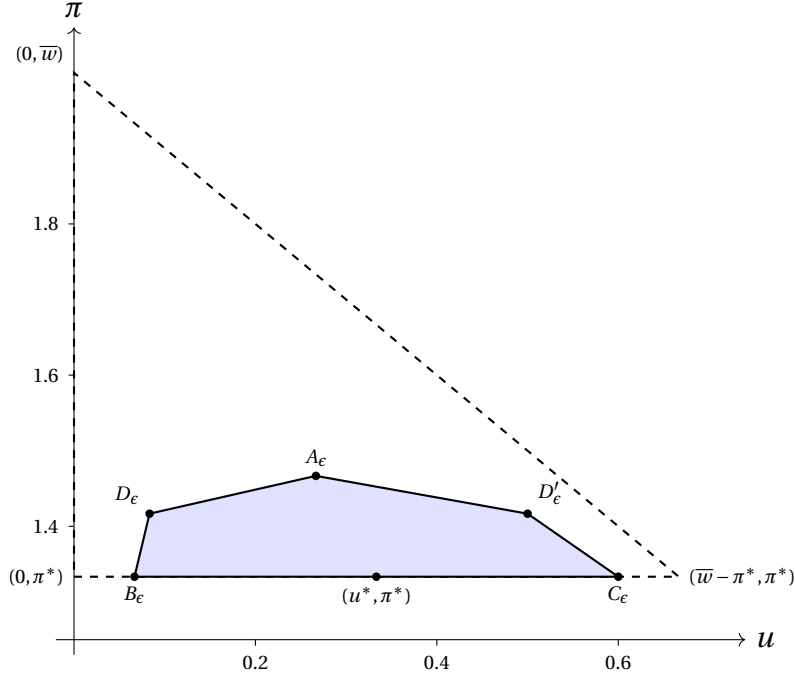


Figure 7: The welfare pentagon for $V = \{1, 2, 3\}$, $x^* = (1/3, 1/3, 1/3)$, and $\epsilon = \log 3$, inside the unrestricted BBM triangle

6. Conclusion

Restrictions on consumer information affect both the markups a firm can charge and the discounts it is willing to offer. We study these two forces by characterizing the consumer and producer surplus attainable under differential privacy. The guarantee determines which price changes disclosure can support; the design of disclosure determines how the market uses those possibilities.

The firm's pricing incentives explain how consumers can benefit despite its option to ignore information. Consumer-optimal disclosure makes the firm just willing to offer discounts: ad-

ditional sales offset the reduction in payments from existing buyers. Every segmentation that maximizes consumer surplus therefore leaves the firm earning its uniform monopoly profit. Under the upper-tail revenue bound, some segmentation also minimizes consumer surplus at this profit level. In that case, consumers' aggregate loss equals the value of trades that no longer occur.

Stronger privacy can protect consumers from higher prices, but it can also prevent discounts and limit the firm's profit gains. With sufficiently strict privacy, the original uniform price remains uniquely optimal after every report, leaving welfare unchanged. Relaxing privacy allows a wider range of outcomes. How much consumers and the firm benefit within this range depends on how disclosure is designed.

Two features of the model suggest further questions. We do not consider consumers' intrinsic preferences for privacy. Evaluating the guarantee itself requires weighing its product-market consequences against the value of privacy, as in the accuracy–privacy tradeoff studied by [Abowd and Schmutte \(2019\)](#). We apply the same privacy rule regardless of how much consumers differ in their willingness to pay. An extension could be protecting small differences more strongly while allowing the firm to distinguish consumers whose valuations are far apart.

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Appendix

A. Proofs and supporting results

A.1 Supporting results for the model

Implementation of private segmentations. Given $\sigma \in \Sigma(\epsilon)$, take $M = \text{supp}(\sigma)$ and define

$$\psi(x \mid v_k) = \frac{\sigma(x) x_k}{x_k^*}.$$

Bayes plausibility gives $\sum_x \psi(x \mid v_k) = 1$. Message x occurs with probability $\sigma(x)$ and induces posterior x . Since

$$\log \frac{\psi(x \mid v_{k'})}{\psi(x \mid v_k)} = \log \frac{x_{k'}}{x_k} - \log \frac{x_{k'}^*}{x_k^*},$$

condition (1) gives ϵ -DP. Conversely, Bayes' rule implies that every ϵ -DP channel induces a segmentation in $\Sigma(\epsilon)$.

Lemma A.1 (Privacy budgets). $X(0) = \{x^*\}$, and $X(\epsilon_1) \subseteq X(\epsilon_2)$ whenever $\epsilon_1 \leq \epsilon_2$.

Proof. At zero budget, (1) makes x proportional to x^* ; normalization gives $x = x^*$. Increasing ϵ weakens every constraint. $\square$

Lemma A.2 (Unique uniform price). Under Assumption 1, $R_{i^*}(x^*) > R_k(x^*)$ for every $k \neq i^*$.

Proof. For $k \neq i^*$, $R_{i^*} \geq R_k$ holds on X^* and is strict at the market concentrated on v^* . Its equality set is therefore a proper face of X^* and cannot contain an interior point. $\square$

Lemma A.3 (Feasible price regions). For finite ϵ , $X(\epsilon)$ and each nonempty $X_k(\epsilon)$ are polytopes consisting of full-support markets.

Proof. The inequalities $x_i x_j^* \leq e^\epsilon x_j x_i^*$ define a bounded intersection of finitely many closed half-spaces within the simplex. A zero coordinate forces all coordinates to vanish, so every

feasible market has full support. Intersecting with the linear inequalities $R_k \geq R_\ell$ gives $X_k(\epsilon)$. $\square$

Proof of Proposition 1. Let $\hat{\pi} = \max_{k \neq i^*} R_k(x^*) < \pi^*$ and $\underline{\epsilon} = \log(\pi^*/\hat{\pi})$. For $x \in X(\epsilon)$, set $a = \min_j x_j/x_j^*$ and $b = \max_j x_j/x_j^*$. Since $b/a \leq e^\epsilon$,

$$\frac{R_k(x)}{R_{i^*}(x)} \leq \frac{bR_k(x^*)}{a\pi^*} \leq e^\epsilon \frac{\hat{\pi}}{\pi^*} < 1 \quad (k \neq i^*, \epsilon < \underline{\epsilon}).$$

Every segment therefore has the unique optimal price v^* . Linearity of R_{i^*} and U_{i^*} and Bayes plausibility give $(u, \pi) = (u^*, \pi^*)$. $\square$

A.2 The binary benchmark

Proof of Proposition 2. Let $\ell = \alpha/(\alpha + e^\epsilon)$, $h = \alpha e^\epsilon/(1 + \alpha e^\epsilon)$, and $c = (v_2 - v_1)/v_2$. At $x_1 < c$, the unique optimal price is v_2 , giving $(u, \pi) = (0, v_2(1 - x_1))$. At $x_1 > c$, it is v_1 , giving $((v_2 - v_1)(1 - x_1), v_1)$. Both prices are optimal at c , and randomization fills the interval between their surplus pairs.

If $\epsilon < \epsilon_B$, $[\ell, h]$ lies strictly on one side of c . Both surpluses are affine there, so Bayes plausibility fixes their expectations at (u^*, π^*) .

For $\epsilon > \epsilon_B$, $\ell < c < h$. The segment-level surplus pairs are affine on each side of c , so their lifted convex hull is generated by

$$\begin{aligned} P_\ell &= (\ell, 0, v_2(1 - \ell)), & P_c^H &= (c, 0, v_1), \\ P_c^L &= (c, (v_2 - v_1)(1 - c), v_1), & P_h &= (h, (v_2 - v_1)(1 - h), v_1), \end{aligned}$$

with coordinates (x_1, u, π) . Feasible outcomes are the slice at $x_1 = x_1^*$. For any two of these points with first coordinates $s < x_1^* < t$, the unique Bayes-plausible weights are

$$\lambda_s = \frac{t - x_1^*}{t - s}, \quad \lambda_t = \frac{x_1^* - s}{t - s}.$$

If $x_1^* < c$, the pairs (P_ℓ, P_h) , (P_ℓ, P_c^H) , and (P_ℓ, P_c^L) give A_ϵ , B_ϵ , and C_ϵ , respectively. If $x_1^* > c$, the corresponding pairs are (P_ℓ, P_h) , (P_c^H, P_h) , and (P_c^L, P_h) . Substituting these weights gives (3)–(6).

These pairs exhaust the slice vertices. Indeed, the mixing weights satisfy two equations: they sum to one and have mean x_1^* . An extreme feasible weight vector has at most two positive entries; with three or more, a nonzero perturbation preserves both equations and remains feasible in both directions. Since none of the four first coordinates equals x_1^* , every such pair must straddle x_1^* . The pairs listed above are the only ones that do. Their welfare images generate the entire slice. Above the threshold, $\pi_A^\epsilon > \pi^*$ and $u_B^\epsilon < u_C^\epsilon$, so all three points are vertices. At $\epsilon = \epsilon_B$, either $h = c$ and $P_h = P_c^L$, or $\ell = c$ and $P_\ell = P_c^H$. The same weights then give the stated segment and the coincident vertices.

For the total-surplus comparison, suppose $\epsilon > \epsilon_B$. High-valuation consumers always buy. If $v^* = v_1$, C_ϵ serves everyone, whereas A_ϵ excludes a positive mass of low-valuation consumers. If $v^* = v_2$, the masses of low-valuation consumers served at C_ϵ and A_ϵ are, respectively,

$$\frac{c(x_1^* - \ell)}{c - \ell} \quad \text{and} \quad \frac{h(x_1^* - \ell)}{h - \ell}.$$

The first is larger because $0 < \ell < c < h$. Thus $w(C_\epsilon) > w(A_\epsilon)$ in either case. Also, B_ϵ and C_ϵ have the same profit and $u_B^\epsilon < u_C^\epsilon$, so C_ϵ uniquely maximizes total surplus. At any finite budget, full support rules out $(0, \bar{w})$: zero consumer surplus requires price v_2 in every segment, excluding all low-valuation consumers. Hence $\mathcal{W}(\epsilon)$ is a strict subset of the unrestricted triangle. $\square$

Proof of Corollary 1. Differentiating (3) gives

$$\frac{du_A^\epsilon}{d\epsilon} = -\frac{(v_2 - v_1)e^\epsilon}{(1 + \alpha)(1 + e^\epsilon)^2} < 0, \quad \frac{d\pi_A^\epsilon}{d\epsilon} = \frac{(v_2 - v_1 + \alpha v_1)e^\epsilon}{(1 + \alpha)(1 + e^\epsilon)^2} > 0.$$

If $v^* = v_2$, $v_2 - v_1 > \alpha v_1$, and (5) fixes B_ϵ and gives

$$\frac{du_C^\epsilon}{d\epsilon} = \frac{\alpha v_1(v_2 - v_1)(v_2 - v_1 - \alpha v_1)e^\epsilon}{(1 + \alpha)((v_2 - v_1)e^\epsilon - \alpha v_1)^2} > 0.$$

If $v^* = v_1$, the inequality reverses, and (6) fixes C_ϵ and gives

$$\frac{du_B^\epsilon}{d\epsilon} = -\frac{\alpha v_1(v_2 - v_1)(\alpha v_1 - (v_2 - v_1))e^\epsilon}{(1 + \alpha)(\alpha v_1 e^\epsilon - (v_2 - v_1))^2} < 0.$$

Equation (4) gives $dw(A_\epsilon)/d\epsilon > 0$. Taking limits in the vertex formulas yields the unrestricted triangle. $\square$

A.3 Feasible markets and welfare outcomes

Proof of Proposition 3. Given a feasible segmentation and optimal pricing rule, collect consumers by the price they face:

$$q_{kj} = \sum_{x \in \text{supp}(\sigma)} \sigma(x) \phi_k(x) x_j.$$

Bayes plausibility gives (7). Whenever $\phi_k(x) > 0$, $x \in X_k(\epsilon)$; summing its privacy and pricing inequalities with weights $\sigma(x)\phi_k(x)$ gives (8) and (9). The surplus formulas follow from the same summation.

Conversely, for each nonzero row of $q \in Q(\epsilon)$, set $s_k = \sum_j q_{kj}$ and $x_j^k = q_{kj}/s_k$. The constraints give $x^k \in X_k(\epsilon)$ and $\sum_k s_k x^k = x^*$. Assign weight s_k and price v_k to this market. If several rows give the same market, combine their weights and randomize over their prices in those proportions. This implements $(u(q), \pi(q))$ using at most one market–price pair per price. Finally, $0 \leq q_{kj} \leq x_j^*$, so $Q(\epsilon)$ is a compact polytope. Its linear welfare image is a compact convex polygon, possibly degenerate. $\square$

Lemma A.4 (The unrestricted limit). *The sets $\mathcal{W}(\epsilon)$ expand with ϵ and are strict subsets of the BBM triangle at every finite budget. Their limit satisfies*

$$\overline{\bigcup_{\epsilon < \infty} \mathcal{W}(\epsilon)} = \{(u, \pi) : u \geq 0, \pi \geq \pi^*, u + \pi \leq \bar{w}\}.$$

Proof of Lemma A.4. Increasing ϵ weakens (8), so $Q(\epsilon)$ and its welfare image are nested. At a finite budget, every nonzero row has full support. Zero consumer surplus then requires price v_K in every row, excluding all lower valuations. Thus $(0, \bar{w})$ is infeasible, and the inclusion in the

BBM triangle is strict.

Let $Q(\infty)$ retain Bayes plausibility and optimal pricing but omit privacy. For each $k \neq i^*$, choose a full-support market close enough to the market concentrated on v_k that v_k remains strictly optimal. Give these markets small positive weights. The residual market is then close to x^* , so it has full support and v^* remains strictly optimal. This constructs $q^0 \in Q(\infty)$ with every entry positive. For any $q \in Q(\infty)$,

$$q^\delta = (1 - \delta)q + \delta q^0, \quad 0 < \delta < 1,$$

remains in $Q(\infty)$ and satisfies privacy for some finite budget because all its entries are positive. As $\delta \downarrow 0$, its welfare converges to that of q . Every unrestricted outcome is therefore a limit of outcomes under finite budgets; Theorem 0 gives the reverse inclusion. $\square$

Lemma A.5 (Privacy vertices). *For $\epsilon > 0$, the vertices of $X(\epsilon)$ are exactly the $2^K - 2$ markets $z^S(\epsilon)$ in (19).*

Proof of Lemma A.5. Write $E = e^\epsilon > 1$ and $\eta_j = x_j / x_j^*$. Feasibility requires $\sum_j x_j^* \eta_j = 1$ and $\eta_j \leq E\eta_\ell$ for all j, ℓ , with every $\eta_j > 0$. Let $m = \min_j \eta_j$ and $M = \max_j \eta_j$. If $M < Em$, all privacy inequalities have strict slack, so the point is not a vertex.

If $M = Em$, let H and L contain the coordinates attaining M and m , and let I contain the others. A privacy inequality binds exactly when its left coordinate belongs to H and its right coordinate belongs to L : equality in $\eta_j \leq M = Em \leq E\eta_\ell$ forces both conditions. The active equalities therefore impose one common low multiplier, a high multiplier E times as large, and leave the $|I|$ intermediate multipliers free. Normalization removes one degree of freedom, since the coefficient on the common low multiplier is

$$E \sum_{j \in H} x_j^* + \sum_{j \in L} x_j^* > 0.$$

If $I \neq \emptyset$, a nonzero direction remains. Small perturbations in both signs preserve normalization, positivity, the active equalities, and all inactive inequalities by their strict slack. The point is not a vertex. If $I = \emptyset$, the active equalities and normalization uniquely determine the point, making

it a vertex. With $S = H$, normalization gives

$$m = \frac{1}{1 + (E - 1)x^*(S)},$$

hence the point is $z^S(\epsilon)$. Every nonempty proper S yields a distinct feasible vertex. $\square$

Let $\mathcal{V}_k(\epsilon)$ denote the vertices of $X_k(\epsilon)$ and $\mathcal{V}(\epsilon) = \bigcup_k \mathcal{V}_k(\epsilon)$. These *relative extreme points* are either privacy vertices or intersections at which price-indifference constraints bind. Linearity of U_k and R_k on $X_k(\epsilon)$ gives

$$\mathcal{W}(\epsilon) = \left\{ (u, \pi) : (x^*, u, \pi) \in \text{co} \bigcup_k \{(x, U_k(x), R_k(x)) : x \in \mathcal{V}_k(\epsilon)\} \right\}. \quad (21)$$

Lemma A.6 (Generating welfare vertices). *Every vertex of $\mathcal{W}(\epsilon)$ is generated by at most K market-price pairs with markets in $\mathcal{V}(\epsilon)$. They can be chosen with linearly independent market vectors, so their weights are unique.*

Proof of Lemma A.6. Choose a linear objective $au + b\pi$ uniquely maximized at the given welfare vertex. By (21), maximize it over nonnegative weights on the finite collection of pairs (x, v_k) with $x \in \mathcal{V}_k(\epsilon)$, subject to

$$\sum_k \sum_{x \in \mathcal{V}_k(\epsilon)} \lambda_{kx} x = x^*.$$

These K equations also imply that the weights sum to one. A basic optimal solution selects at most K linearly independent market columns and must give the chosen welfare vertex. Bayes plausibility uniquely determines the selected weights. $\square$

Unlike the direct representation, this vertex representation may use several markets for one price: pooling them preserves welfare but may move the posterior away from a relative extreme point.

A.4 Consumer-surplus endpoints

Write $p = x^*$ and $i = i^*$. For unnormalized market masses, let

$$D(\epsilon) = \{y \geq 0 : y_j p_\ell \leq e^\epsilon y_\ell p_j \text{ for all } j, \ell\},$$

and let $C_k(\epsilon) = \{y \in D(\epsilon) : R_k(y) \geq R_\ell(y) \text{ for all } \ell\}$. Both are closed convex cones. Every nonzero private row has full support. Privacy bounds its largest likelihood multiplier by e^ϵ times its smallest; keeping all multipliers within their original range therefore preserves privacy. The profit-preserving arrays form

$$Q^*(\epsilon) = \{q \in Q(\epsilon) : R_k(q_k) = R_i(q_k) \text{ for every } k\}. \quad (22)$$

Proof of Theorem 1. Bayes plausibility gives

$$\pi(\sigma, \phi) - \pi^* = \sum_x \sigma(x) [\max_k R_k(x) - R_i(x)].$$

Each summand is nonnegative, so equality holds exactly when v_i is optimal in every positive-weight segment. In direct form, these arrays constitute $Q^*(\epsilon)$, a nonempty compact convex face of $Q(\epsilon)$. Its linear image under u is the interval $[u^-, u^+]$.

For any nonzero private row y and $k < m$,

$$U_k(y) - U_m(y) = \sum_{j=k}^{m-1} (v_j - v_k) y_j + (v_m - v_k) \sum_{j=m}^K y_j > 0.$$

Hence a floor endpoint must select the highest tied price for the minimum and the lowest for the maximum; otherwise, repricing the same segmentation improves consumer surplus while preserving profit. $\square$

Lemma A.7 (Prices at consumer-surplus extrema). *Every q attaining $u_{\min}(\epsilon)$ has $q_k = 0$ for $k < i^*$. Every q attaining $u_{\max}(\epsilon)$ has $q_k = 0$ for $k > i^*$.*

Proof. Put $H = \sum_{j \geq i} p_j$ and $H_r = \sum_{j \geq i} q_r p_j$. The two relaxations below retain Bayes plausibility

and privacy. Both are compact and contain the uninformative array.

For the minimum, set rows below i to zero and require only $q_i \in D(\epsilon)$; higher rows remain obedient. Pooling all rows $k \leq i$ into row i satisfies this relaxation and weakly lowers consumer surplus, strictly if a lower row is nonzero.

Take a relaxed minimizer. Keep its tail cells and set

$$\hat{q}_{rj} = p_j H_r / H, \quad j < i.$$

Column sums and consumer surplus are unchanged. Each new multiplier H_r / H averages the unchanged tail multipliers, so privacy is preserved. Every nonzero row satisfies

$$R_\ell(\hat{q}_r) = \frac{H_r}{H} R_\ell(p) < \frac{H_r}{H} R_i(p) = R_i(\hat{q}_r), \quad \ell < i.$$

Higher rows remain obedient because their tail revenues are unchanged. If row i preferred a higher price, repricing and merging it into that price's row would be feasible and strictly lower consumer surplus, contradicting the relaxed minimum. Thus the relaxation is exact. The strict initial pooling inequality excludes lower rows from every original minimizer.

For the maximum, set rows above i to zero, retain obedience below i , and require only $q_i \in D(\epsilon)$. Pooling rows $k \geq i$ into row i weakly raises consumer surplus, strictly if a higher row is nonzero. In a relaxed maximizer, keep the cells below i and set

$$\hat{q}_{rj} = p_j H_r / H, \quad j \geq i.$$

This averaging preserves privacy, column sums, and consumer surplus, since

$$u(q) = \sum_{r \leq i} \sum_{j=r}^{i-1} (v_j - v_r) q_{rj} + \sum_{j \geq i} v_j p_j - \sum_{r \leq i} v_r H_r.$$

Revenues at prices at most v_i are unchanged, while every nonzero row satisfies

$$R_h(\hat{q}_r) = \frac{H_r}{H} R_h(p) < \frac{H_r}{H} R_i(p) = R_i(\hat{q}_r), \quad h > i.$$

If row i preferred a lower price, repricing and merging it there would improve on the relaxed maximum. Hence this relaxation is also exact, and no original maximizer uses a higher row. $\square$

Lemma A.8 (Pooling at an extremum). *At a consumer-surplus minimum, the firm charges the highest optimal price in each segment. If positive segments x, x' receive prices $v_k < v_m$, some $v_h < v_m$ is also optimal in x' and satisfies $R_h(x) > R_m(x)$.*

At a consumer-surplus maximum, the firm charges the lowest optimal price. For segments priced at $v_k < v_m$, some $v_h > v_k$ is optimal in the lower-priced segment and satisfies $R_h(x') > R_k(x')$.

Proof. Strict monotonicity of U_k gives the tie-breaking claims. At a minimum, suppose every lower price tied with v_m in x' satisfied $R_h(x) \leq R_m(x)$. Then v_m remains optimal after adding a sufficiently small amount of x to x' : tied lower prices do not overtake it, and all other prices have strict slack, including higher prices by tie-breaking. Convexity preserves privacy, and transferring mass between these segments preserves Bayes plausibility. Charging the transferred consumers v_m strictly lowers their surplus, a contradiction. At a maximum, the same argument transfers mass from x' to x ; tied higher prices cannot overtake v_k unless one satisfies the stated inequality, and all other prices have strict slack. $\square$

Lemma A.9 (Directional reduction). *Let $H = \sum_{j=i}^K p_j$. The tail market*

$$V_T = (v_i, \dots, v_K), \quad p_T = (p_i/H, \dots, p_K/H)$$

has unique monopoly price v_i and satisfies

$$u_{\min}(\epsilon) = H u_{\min, T}(\epsilon), \quad u^-(\epsilon) = H u_T^-(\epsilon).$$

The collapsed market

$$V_C = (v_1, \dots, v_i), \quad p_C = (p_1, \dots, p_{i-1}, H)$$

also has unique monopoly price v_i and, with $B = \sum_{j \geq i} (v_j - v_i) p_j$, satisfies

$$u_{\max}(\epsilon) = B + u_{\max, C}(\epsilon), \quad u^+(\epsilon) = B + u_C^+(\epsilon).$$

Both reductions keep the same privacy budget. Their active prices are, respectively, the original active prices at or above v_i and at or below v_i .

Proof of Lemma A.9. The tail market scales aggregate revenues at prices $v_r \geq v_i$ by $1/H$; the collapsed market leaves revenues at prices $v_r \leq v_i$ unchanged. Both retain the unique monopoly price v_i .

By Lemma A.7, a consumer-surplus minimizer uses only rows $r \geq i$. For any feasible array with this support, the restriction

$$\bar{q}_{rj} = q_{rj}/H, \quad r, j \geq i,$$

preserves Bayes plausibility and obedience in the tail market. Likelihood multipliers are unchanged, so privacy holds, and $u(q) = Hu_T(\bar{q})$.

Conversely, given a feasible tail array $\bar{q}$, let $s_r = \sum_{j=i}^K \bar{q}_{rj}$ and define

$$q_{rj} = H\bar{q}_{rj} \quad (j \geq i), \quad q_{rj} = p_j s_r \quad (j < i), \quad r \geq i,$$

with zero rows below i . Column sums equal p , and each added multiplier s_r averages the row's tail multipliers. Thus privacy holds. The added prices are strictly worse in every nonzero row:

$$R_\ell(q_r) = s_r R_\ell(p) < s_r R_i(p) = R_i(q_r) \leq R_r(q_r), \quad \ell < i.$$

These maps prove $u_{\min} = Hu_{\min, T}$ and preserve profit gaps up to scale:

$$R_r(q_r) - R_i(q_r) = H[R_{r, T}(\bar{q}_r) - R_{i, T}(\bar{q}_r)].$$

The floor minimum also uses only prices at least v_i by Theorem 1, so the same maps give $u^- = Hu_T^-$.

For the maximum, consider a feasible array with zero rows above i and collapse each tail into

$$\tilde{q}_{ri} = \sum_{j=i}^K q_{rj},$$

retaining cells below i . The multiplier $\tilde{q}_{ri}/H$ averages the tail multipliers, so privacy holds. Column sums and revenues at prices at most v_i are preserved, and $u(q) = B + u_C(\tilde{q})$.

To lift a feasible collapsed array, retain its cells below i and set $q_{rj} = (p_j/H)\tilde{q}_{ri}$ for $j \geq i$. Privacy, column sums, and lower-price revenues are preserved; every higher price is strictly worse in a nonzero row:

$$R_h(q_r) = \frac{\tilde{q}_{ri}}{H} R_h(p) < \frac{\tilde{q}_{ri}}{H} R_i(p) = R_i(q_r) \leq R_r(q_r), \quad h > i.$$

Lemma A.7 now gives $u_{\max} = B + u_{\max,C}$. The maps preserve each row's profit gap, and the floor maximum uses only prices at most v_i by Theorem 1; hence $u^+ = B + u_C^+$.

The same restrictions and lifts apply to individual nonzero rows without imposing column sums. They preserve privacy and optimality of the relevant price, so normalization gives both active-price identities. $\square$

Proof of Theorem 2. Lemmas A.7 and A.9 map every consumer-surplus maximizer to one in the collapsed market, preserving producer surplus. It suffices to consider $v^* = v_K$; the case $K = 1$ is immediate. Write p for the reduced prior, $E = e^\epsilon$, and $a_{rj} = q_{rj}/p_j$.

Fix any maximizer $\tilde{q}$. Among maximizers with producer surplus $\pi(\tilde{q})$, choose q to minimize

$$\mathcal{P}(q) = \sum_{r,j} \frac{q_{rj}^2}{p_j}.$$

The constraint set is nonempty and compact. In every nonzero row,

$$R_r(q_r) > R_h(q_r) \quad (h < r). \tag{23}$$

Otherwise, repricing at a tied lower price and merging with that price's row preserves feasibility and strictly raises consumer surplus.

Call $R_k(q_k) - R_K(q_k)$ the profit gap in row k . We prove by descending induction on $j = K-1, \dots, 1$

that every row with a positive gap satisfies

$$a_{kj} \leq a_{kK}. \quad (24)$$

Assume the inequalities for indices above j (an empty hypothesis when $j = K-1$). For a positive-gap row k and $j < h < K$,

$$R_h(q_k) \leq a_{kK} R_h(p) < a_{kK} R_K(p) = R_K(q_k) < R_k(q_k). \quad (25)$$

The prior's unique monopoly price gives the first strict inequality. Full support gives $a_{kK} > 0$. Price ν_K is also strictly worse by the positive gap. Hence $k \leq j$, and every price above ν_j is strictly suboptimal in row k .

Suppose $a_{kj} > a_{kK}$. Since $\sum_r a_{rj} = \sum_r a_{rK} = 1$, another row r satisfies $a_{rj} < a_{rK}$. Change four cells:

$$\begin{aligned} q'_{kj} &= q_{kj} - \delta, & q'_{kK} &= q_{kK} + \delta, \\ q'_{rj} &= q_{rj} + \delta, & q'_{rK} &= q_{rK} - \delta. \end{aligned}$$

Column sums are unchanged. Choose

$$0 < \delta < \frac{\min\{a_{kj} - a_{kK}, a_{rK} - a_{rj}\}}{p_j^{-1} + p_K^{-1}}.$$

In each changed row, the two multipliers move toward one another without crossing. Let $m_t = \min_{\ell} a_{t\ell} > 0$ and $M_t = \max_{\ell} a_{t\ell}$ for $t \in \{k, r\}$. Every new multiplier lies in $[m_t, M_t]$, so all ordered pairs satisfy

$$a'_{t\ell} \leq M_t \leq E m_t \leq E a'_{th}.$$

Thus privacy and nonnegativity hold in both changed rows.

In row k , revenues at prices at most ν_j are unchanged; higher-price revenues increase by $\nu_h \delta$. Their strict slack, established above, preserves obedience for small δ .

In row r , lower-price revenues are unchanged and revenues above v_j fall by $v_h\delta$. If $r \leq j$, obedience follows directly. If $r > j$, write $G_h = R_r(q_r) - R_h(q_r)$. The new gaps are

$$R_r(q'_r) - R_h(q'_r) = \begin{cases} G_h - v_r\delta, & h \leq j, \\ G_h - (v_r - v_h)\delta, & j < h < r, \\ G_h + (v_h - v_r)\delta, & h > r. \end{cases}$$

The first two cases have $G_h > 0$ by (23); the last expression is nonnegative. There are finitely many comparisons, so one sufficiently small positive δ preserves obedience and privacy in both rows.

Since $k \leq j$, consumer surplus changes by

$$u(q') - u(q) = \begin{cases} 0, & r \leq j, \\ (v_r - v_j)\delta > 0, & r > j. \end{cases}$$

The second case contradicts consumer optimality. In the first, both assigned revenues are unchanged, so q' also has producer surplus $\pi(\bar{q})$. But

$$\left. \frac{d\mathcal{P}(q')}{d\delta} \right|_{\delta=0} = 2[(a_{kK} - a_{kj}) + (a_{rj} - a_{rK})] < 0,$$

contradicting the choice of q for sufficiently small δ . This proves (24) and completes the induction.

A positive-gap row $k < K$ would therefore satisfy

$$R_k(q_k) \leq a_{kK}R_k(p) < a_{kK}R_K(p) = R_K(q_k),$$

contrary to obedience. Row K has zero gap, so all rows preserve profit and $\pi(q) = \pi^*$. Because $\pi(q) = \pi(\bar{q})$ and $\bar{q}$ was arbitrary, every maximizer preserves profit. The reduction gives the result in the original market, including $u_{\max}(c) = u^+(c)$. $\square$

Proof of Proposition 4. Lemma A.9 reduces the problem to the tail market with $v^* = v_1$. If no higher price is active, only v_1 can be charged and $u_{\min} = u^- = u^*$. Otherwise, relabel the tail and write p for its prior. The reduction preserves active prices and scales the assumed bound to $v_b \sum_{j=a}^K p_j \leq v_1$.

Among consumer-surplus minimizers, minimize producer surplus, then minimize

$$I(q) = \sum_{r:s_r>0} \sum_j q_{rj} \log \frac{q_{rj}}{p_j s_r}, \quad s_r = \sum_j q_{rj}.$$

These minima exist by compactness: both surpluses are linear, and

$$I(q) = \sum_{r,j} q_{rj} \log(q_{rj}/p_j) - \sum_r s_r \log s_r$$

is continuous with $0 \log 0 = 0$. Write $a_{rj} = q_{rj}/p_j$.

Suppose row k has a positive profit gap, $R_k(q_k) > R_1(q_k)$. We claim

$$a_{kj} \geq s_k \quad (j < a).$$

Otherwise, $\sum_r a_{rj} = \sum_r s_r = 1$ supplies a donor row r with $a_{rj} > s_r$. Transfer a small mass $\delta > 0$ of type j from r to k . The recipient's multiplier rises from below its row average and the donor's falls from above its average. Choose δ so both remain within their original row's multiplier range; this preserves every privacy inequality and nonnegativity. Column sums are unchanged.

Since $k \geq a > j$, recipient revenue is unchanged. Only competitors at or below v_j gain revenue: v_1 has strict slack by the positive gap, and every other such price is inactive, hence strictly suboptimal in any nonzero private row. Small δ preserves obedience.

If the donor charges a price at least v_a , its assigned revenue is also unchanged and competing revenues weakly fall. Both surpluses stay fixed, but

$$\left. \frac{dI}{d\delta} \right|_{\delta=0} = \log \frac{a_{kj}}{s_k} - \log \frac{a_{rj}}{s_r} < 0,$$

contradicting the choice of q . The only other possible donor price is v_1 . It is uniquely optimal in that row by Lemma A.8, so a small transfer again preserves obedience. For $j > 1$, consumer surplus falls by $(v_j - v_1)\delta$; for $j = 1$, it is unchanged and profit falls by $v_1\delta$. Either contradicts the successive minimizations, proving the claim.

The claim implies

$$\sum_{j=a}^K q_{kj} \leq s_k \sum_{j=a}^K p_j.$$

If at least two higher prices are active, every one except the highest has $a \leq h \leq b$, so

$$R_h(q_k) \leq v_b \sum_{j=a}^K q_{kj} \leq v_b s_k \sum_{j=a}^K p_j \leq v_1 s_k = R_1(q_k) < R_k(q_k).$$

Price v_1 and every inactive price are also strictly worse. Thus v_k is the highest active price and is uniquely optimal in row k . With only one higher active price, the positive gap gives this conclusion directly.

Some other nonzero row must charge less than v_k ; otherwise, summing obedience inequalities makes v_k optimal in the prior, contrary to $v^* = v_1$. Lemma A.8 then requires a lower price to tie v_k in row k , contradicting its unique optimality.

Hence every row of the selected minimizer preserves profit, giving $u_{\min} = u^-$. The tail-market lift gives the equality in the original market. $\square$

Proof of Proposition 5. Every nonzero row contains every valuation. Thus zero consumer surplus requires charging v_K in every row. Aggregating obedience inequalities makes v_K optimal in the prior, so $v^* = v_K$. Conversely, the uninformative segmentation attains zero consumer surplus when $v^* = v_K$.

First-best total surplus requires charging v_1 in every row, since any higher price excludes some consumers. Aggregation then gives $v^* = v_1$. Also, $u \leq \bar{w} - \pi^*$, with equality only if total surplus is first-best and profit is π^* . The uninformative segmentation attains both when $v^* = v_1$, proving the total-surplus and maximum-consumer-surplus claims.

Finally, $\pi = \bar{w}$ requires both zero consumer surplus and first-best total surplus, hence every row charging both v_K and v_1 . This is impossible for $K \geq 2$; compactness makes the bound on maximum producer surplus strict. $\square$

A.5 Price-activation thresholds

Lemma A.10 (Computing activation thresholds). *For each k , the minimum in (16) is attained and*

$$e^{\epsilon_k^{\text{act}}} = \min_{\tau, r} \left\{ \tau : \begin{array}{l} 1 \leq r_j \leq \tau \quad \text{for every } j, \\ v_k \sum_{j=k}^K x_j^* r_j \geq v_\ell \sum_{j=\ell}^K x_j^* r_j \quad \text{for every } \ell \end{array} \right\}. \quad (26)$$

The uniform price has threshold zero; all other thresholds are positive and finite. Price v_k is active exactly when $\epsilon \geq \epsilon_k^{\text{act}}$.

Proof of Lemma A.10. For any full-support $x \in X_k$, setting

$$r_j = \frac{x_j / x_j^*}{\min_m (x_m / x_m^*)}$$

makes $(e^{\rho(x)}, r)$ feasible in (26). Conversely, a feasible (τ, r) induces

$$x_j = \frac{x_j^* r_j}{\sum_m x_m^* r_m} \in X_k, \quad \rho(x) \leq \log \tau.$$

Thus the two programs have the same infimum after exponentiation. A market sufficiently close to the point mass at v_k , mixed with a positive weight on x^* , has full support and keeps v_k strictly optimal. The linear program is therefore feasible at some finite τ_0 . Its feasible set with $\tau \leq \tau_0$ is compact, so both minima are attained.

Since $\rho(x) = 0$ only at x^* , uniqueness of the uniform price gives $\epsilon_{i^*}^{\text{act}} = 0$ and $\epsilon_k^{\text{act}} > 0$ for $k \neq i^*$. Attainment and (15) give the activation equivalence, including equality at the threshold. $\square$

Proof of Proposition 6. Write $p = x^*$, $i = i^*$, and $E = e^\epsilon$. At a full-support market, $U_k(x)$ strictly decreases with k . If no higher price is active, every feasible outcome has $u \geq u^*$; if no lower

price is active, every outcome has $u \leq u^*$. The uninformative segmentation attains u^* .

Any active nonuniform price can change consumer surplus. Take $x \in X_k(\epsilon)$ with $k \neq i$, so $\epsilon > 0$. At p , all nontrivial privacy inequalities and all revenue comparisons favoring v_i are strict. Hence, for sufficiently small $\lambda > 0$,

$$y_\lambda = \frac{p - \lambda x}{1 - \lambda}$$

has full support, satisfies privacy, and has unique optimal price v_i : it converges to p , and there are only finitely many inequalities to preserve. Give x weight λ at price v_k and y_λ weight $1 - \lambda$ at price v_i . Bayes plausibility then gives

$$u - u^* = \lambda[U_k(x) - U_i(x)],$$

which is negative for $k > i$ and positive for $k < i$. This proves the claims for $u_{\min}$ and $u_{\max}$.

To obtain a loss on the profit floor, take an active higher-price market $x \in X_k(\epsilon)$, $k > i$. Set $H = \sum_{j \geq i} p_j$, $T = \sum_{j \geq i} x_j$, and

$$p_{T,j} = p_j / H, \quad z_j = x_j / T \quad (j \geq i).$$

The likelihood multipliers of z relative to p_T are $(H/T)(x_j/p_j)$, so z satisfies tail-market privacy. Since tail revenues equal the corresponding original revenues divided by T , v_k remains optimal among tail prices.

The path $z(t) = (1 - t)p_T + tz$ stays in the tail privacy polytope by convexity. Define

$$g(t) = \max_{h > i} \{R_{h,T}(z(t)) - v_i\}.$$

The lowest tail price earns v_i throughout. Uniqueness of the uniform price gives $g(0) < 0$, while optimality of v_k at z gives $g(1) \geq 0$. Continuity ensures that

$$t_0 = \min\{t \in [0, 1] : g(t) \geq 0\} \in (0, 1), \quad g(t_0) = 0.$$

At $z(t_0)$, every higher price earns at most v_i and some $v_h > v_i$ earns exactly v_i .

Lift this market by setting

$$\hat{x}_j = \begin{cases} p_j, & j < i, \\ Hz_j(t_0), & j \geq i. \end{cases}$$

The tail multipliers $b_j = z_j(t_0)/p_{T,j}$ average to one. If $m = \min b_j$ and $M = \max b_j$, then $m \leq 1 \leq M \leq Em$. The added lower-type multipliers equal one, so every original multiplier lies in $[m, M]$ and every pair satisfies privacy. Higher-price revenues are H times their tail revenues; lower-price revenues satisfy

$$R_\ell(\hat{x}) = R_\ell(p) < R_i(p) = R_i(\hat{x}), \quad \ell < i.$$

Thus $\hat{x} \in X_i(\epsilon) \cap X_h(\epsilon)$: the tie is optimal against every original price.

Use the residual construction with $\hat{x}$, charging v_h there and v_i in the residual market. Both segments keep v_i optimal, so profit is π^* , while

$$u - u^* = \lambda[U_h(\hat{x}) - U_i(\hat{x})] < 0.$$

Hence an active higher price implies $u^- < u^*$. The converse follows from $u^- \geq u_{\min}$. The threshold formulas now follow from (17) and (18); Theorem 2 also gives $u^+ = u_{\max}$. $\square$

A.6 Producer-optimal disclosure

Proof of Proposition 7. At $\epsilon = 0$, all $z^S(0) = x^*$ and one market suffices. For $\epsilon > 0$, monopoly profit $f(x) = \max_k R_k(x)$ is convex. By Lemma A.5, write each private market as $x = \sum_S \mu_x(S) z^S(\epsilon)$. Replacing that segment with its vertex decomposition preserves Bayes plausibility and weakly raises profit:

$$f(x) \leq \sum_S \mu_x(S) f(z^S(\epsilon)).$$

The stated linear program therefore attains maximum producer surplus. Summing its Bayes-plausibility equations gives $\sum_S \lambda_S = 1$, so its feasible set is compact. A basic optimal solution uses at most K markets, the number of equations.

The upper boundary of a compact convex polygon is concave and piecewise linear, with finitely many kinks generated as in Lemma A.6. If $\bar{\pi}_\epsilon(u_A) > \pi^*$ somewhere, mix that point with an outcome at the opposite endpoint of the consumer-surplus range to reach any other interior u . Every feasible outcome has profit at least π^* , so this mixture has profit strictly above π^* . In the binary case, Proposition 2 gives the unique kink A_ϵ above the activation threshold. $\square$

A.7 The three-valuation example

Return to $V = \{1, 2, 3\}$, $x^* = (1/3, 1/3, 1/3)$, and write $E = e^\epsilon$. Both discriminatory prices activate at $E = 2$. For $E > 2$, the welfare polygon has vertices

$$\begin{aligned} B_\epsilon &= \left(\frac{1}{3(2E-1)}, \frac{4}{3} \right), & C_\epsilon &= \left(\frac{4E-3}{3(2E-1)}, \frac{4}{3} \right), \\ D_\epsilon &= \left(\frac{1}{3(E+1)}, \frac{5E+2}{3(E+1)} \right), & D'_\epsilon &= \left(\frac{E+3}{3(E+1)}, \frac{5E+2}{3(E+1)} \right), \\ A_\epsilon &= \left(\frac{4}{3(E+2)}, \frac{6E+4}{3(E+2)} \right). \end{aligned}$$

Write the rows of a feasible assignment as $a_j = q_{1j}$, $b_j = q_{2j}$, and $c_j = q_{3j}$. Thus $a_j + b_j + c_j = 1/3$, and

$$u = a_2 + 2a_3 + b_3, \quad \pi = a_1 + a_2 + a_3 + 2b_2 + 2b_3 + 3c_3.$$

We show that the welfare set is exactly the intersection of the half-planes

$$\pi \geq \frac{4}{3}, \quad (27)$$

$$\pi - (2E - 1)u \leq 1, \quad (28)$$

$$(3E + 2)\pi - E(E - 2)u \leq \frac{14E + 4}{3}, \quad (29)$$

$$(E^2 + E + 2)\pi + E(E - 2)u \leq \frac{6E^2 + 2E + 4}{3}, \quad (30)$$

$$2E\pi + (2E - 1)u \leq 4E - 1. \quad (31)$$

Optimality against price 2 gives the profit floor:

$$\pi - \frac{4}{3} = (a_1 - a_2 - a_3) + (c_3 - 2c_2) \geq 0.$$

For the remaining inequalities, set

$$d = \frac{E + 2}{E(E - 1)}, \quad \kappa = \frac{E^2 - 4}{E - 1}.$$

For $E \geq 2$, we have $0 < d \leq 2$ and $\kappa \geq 0$. Bayes plausibility and the surplus formulas give the following identities for the four upper-facet slacks:

$$\begin{aligned}
& 1 - \pi + (2E - 1)u \\
&= (Ea_3 - a_1) + (2E - 2)a_2 + 3Ea_3 + 2(Eb_3 - b_2), \\
&\frac{14E + 4}{3} - (3E + 2)\pi + E(E - 2)u \\
&= (E + d)(Ea_2 - a_1) + (2E + 2 - d)(Ea_3 - a_1) \\
&\quad + \kappa(Eb_3 - b_2) + \left(5 + \frac{3}{E - 1}\right)(Ec_2 - c_3), \\
&\frac{6E^2 + 2E + 4}{3} - (E^2 + E + 2)\pi - E(E - 2)u \\
&= (2 + d)(Ea_2 - a_1) + \left(3 + \frac{2}{E}\right)(Ea_3 - a_1) \\
&\quad + \kappa(Eb_1 - b_2) + \kappa(Ec_1 - c_3) + E(2 + d)(Ec_2 - c_3), \\
&4E - 1 - 2E\pi - (2E - 1)u \\
&= (Eb_1 - b_2) + (Eb_1 - b_3) + 2(Ec_1 - c_3) + (4E - 1)c_2.
\end{aligned}$$

Privacy and nonnegativity make every term on the right nonnegative, proving (27)–(31).

For $E > 2$, the four upper boundary lines have strictly decreasing slopes

$$2E - 1 > \frac{E(E - 2)}{3E + 2} > 0 > -\frac{E(E - 2)}{E^2 + E + 2} > -\frac{2E - 1}{2E}.$$

The consecutive intersections, including those with the profit floor, are B_e, D_e, A_e, D'_e, C_e , in strictly increasing order of consumer surplus. The five half-planes therefore intersect in exactly the proposed pentagon.

The following assignments attain its vertices; rows correspond to prices 1, 2, and 3:

$$\begin{aligned}
q^B &= \frac{1}{3(2E-1)} \begin{pmatrix} 0 & 0 & 0 \\ E & E & 1 \\ E-1 & E-1 & 2(E-1) \end{pmatrix}, & q^C &= \frac{1}{3(2E-1)} \begin{pmatrix} 2(E-1) & E-1 & E-1 \\ 1 & E & E \\ 0 & 0 & 0 \end{pmatrix}, \\
q^D &= \frac{1}{3(E+1)} \begin{pmatrix} 0 & 0 & 0 \\ E & E & 1 \\ 1 & 1 & E \end{pmatrix}, & q^{D'} &= \frac{1}{3(E+1)} \begin{pmatrix} E & 1 & 1 \\ 1 & E & E \\ 0 & 0 & 0 \end{pmatrix}, \\
q^A &= \frac{1}{3(E+2)} \begin{pmatrix} E & 1 & 1 \\ 1 & E & 1 \\ 1 & 1 & E \end{pmatrix}.
\end{aligned}$$

Every column sums to $1/3$, and each nonzero row has maximum-to-minimum ratio E or 2 , so privacy holds for $E \geq 2$. Optimality at price 1 requires $a_1 \geq a_2 + a_3$ and $a_1 + a_2 \geq 2a_3$; at price 2, it requires $b_2 + b_3 \geq b_1$ and $2b_2 \geq b_3$; and at price 3, it requires $2c_3 \geq c_1 + c_2$ and $c_3 \geq 2c_2$. The displayed assignments satisfy these inequalities for $E \geq 2$ and yield the stated surplus pairs. Convexity makes the whole pentagon attainable.

In q^B , the price-3 row has posterior $(1/4, 1/4, 1/2)$ and weight $4(E-1)/(3(2E-1))$; the remaining row is $z^{\{1,2\}}(\epsilon)$ at price 2. In q^C , the same weight is placed on $(1/2, 1/4, 1/4)$ at price 1, with the remainder on $z^{\{2,3\}}(\epsilon)$ at price 2. At each indifference market the nonuniform price ties 2. The other assignments use only privacy vertices.

At $E = 2$, the same valid inequalities impose $\pi = 4/3$ and $1/9 \leq u \leq 5/9$; q^B and q^C attain the endpoints. For $E < 2$, a nonzero price-1 row would require $a_1 \geq a_2 + a_3 \geq 2 \min\{a_2, a_3\}$, contrary to $a_1 \leq E \min\{a_2, a_3\}$. Likewise, a nonzero price-3 row would require $c_3 \geq 2c_2$, contrary to $c_3 \leq Ec_2$. Hence only price 2 can be charged and the welfare set is $\{(1/3, 4/3)\}$. As $\epsilon \rightarrow \infty$, B_ϵ , C_ϵ , and A_ϵ approach the vertices of the BBM triangle, while D_ϵ and D'_ϵ approach $(0, 5/3)$ and $(1/3, 5/3)$ on its edges.